Time-Optimal Candidate Heliocentric Transfers With a Constant-Power, Variable-I_{sp} Engine

Jan Olšina

Independent Researcher, Prague, Czech Republic
email: jan.olsina@gmail.com, ORCID: 0009-0003-9816-3958.

***Abstract*—We study time-optimal candidate heliocentric transfers for a spacecraft propelled by an electric thruster that supplies constant useful jet power P while continuously varying its exhaust speed (variable I_{sp}). The vehicle is assumed to depart from and arrive on circular heliocentric orbits, so that the initial and final velocities match the local circular velocities at the respective radii. First, we derive the analytic one-dimensional, gravity-free brachistochrone and characterize the normal-PMP structure produced by a finite exhaust-speed ceiling. The ceiling introduces maximum-exhaust-speed boundary arcs and a central coast; every finite-switch candidate is an endpoint truncation of an interior-boost–boundary-boost–coast–boundary-brake–interior-brake–boundary-brake sequence. On boundary arcs the mass costate is not conserved, and the braking branch cannot return to the coast threshold. Next, we formulate the full planar Sun-field optimal-control problem, derive two closed-form first integrals, and show that the indirect formulation reduces the orbital/costate shooting dynamics to seven differential equations, with propellant consumption enforced by an additional quadrature or endpoint constraint. Finally, we present a practical numerical continuation strategy that searches for a coarse endpoint via global optimization and then refines it by homotopy and Powell's local solver. Ideal uncapped numerical examples for a $1\,\mathrm{GW}$ useful jet-power engine with an initial/dry mass of $3000\,\mathrm{t} \to 1000\,\mathrm{t}$ produce Earth–Jupiter-class ideal transfer candidates whose final-angle table spans flight times ranging from roughly 170 days up to about 220 days; the longer-arc candidates highlighted in the figures commonly exploit a low-perihelion Oberth-like pass. An audit verifies endpoint, dry-mass and free-time normalization conditions for 73 reconstructed Jupiter rows and reports their radius and exhaust diagnostics. All 73 leave a 50–250 km/s exhaust band, so the tables are ideal-engine benchmarks rather than missions for a bounded engine.**

I. Introduction

While electric propulsion (EP) is traditionally valued for its propellant efficiency, future ambitious endeavors such as interplanetary logistics networks or rapid sample return missions will demand not just efficiency but *speed* [1], [2], [3]. For such missions, the design focus shifts from optimizing propellant mass to minimizing flight time, making the spacecraft's power system the primary limiting factor [4]. Reducing transfer times from years to months propels interest in high-power propulsion systems, with gigawatt-class architectures representing a plausible long-term goal. Several pathways to this power scale are being explored, including advanced Nuclear Electric Propulsion (NEP) [5], [6], next-generation deployable solar arrays [7], and beamed-energy concepts [8]. A common feature of these technologies is the ability to provide nearly constant useful propulsion power over long thrust arcs. This constant-power paradigm fundamentally alters the classic rocketry framework. Instead of the fixed exhaust velocity (v_e) of the Tsiolkovsky equation [9], the exhaust speed becomes a key control variable, allowing the engine to actively trade thrust for specific impulse. The Variable Specific Impulse Magnetoplasma Rocket (VASIMR) is an exemplary technology demonstrating this "constant-power throttling" over a wide operating range [10]. Such throttleability is compelling for the fast transfers that motivate our study, as recent optimization work confirms that varying the specific impulse during flight can significantly improve trip time [11]. In this paper, we develop the analytical and numerical machinery for constructing time-optimal transfer candidates in this high-power regime. We first derive an analytical solution for a 1D, gravity-free "brachistochrone" to build intuition. We then formulate the full planar Sun-field problem, derive two closed-form first integrals to simplify the dynamics, and present a robust numerical strategy to solve the resulting boundary-value problem. Our idealized uncapped examples for a gigawatt-scale engine demonstrate Earth–Jupiter-class transfer candidates in a few months, frequently exploiting a low-perihelion Oberth-like maneuver. The finite-ceiling analysis is used to characterize the limitation of that benchmark rather than in the reported heliocentric computations.

II. Variable-I_{sp} Rocket Model

Let $m(t)$ be the instantaneous mass ($m \geq m_{\mathrm{dry}}$) and $v_e(t)$ the exhaust speed. The propellant mass flow rate is the positive quantity $\dot{m}_f = -\dot{m}$. Neglecting plume inefficiencies, the useful jet power P (taken to be constant) and thrust magnitude T are given by [12]:

$$P = \tfrac{1}{2}(-\dot{m})\, v_e^2, \qquad T = m\,a = (-\dot{m})\, v_e, \tag{1}$$

where a is the resulting thrust-induced acceleration magnitude.[1] Eliminating v_e gives a direct relationship between the mass flow rate and the vehicle's acceleration:

$$\dot{m} = -\frac{m^2 a^2}{2P}. \tag{2}$$

This equation forms the basis of our optimal control problem.

[1] In the vector formulation (Sec. V), $a = \|\mathbf{a}_T\|$ is the thrust-acceleration magnitude. In the free-space derivation below $a(t)$ is used as a signed scalar; since (2) depends on a^2, the sign convention is immaterial for the propellant balance.

III. Exact Solution for the Free-Space Case ($\mu = 0$)

With solar gravity absent ($\mu = 0$), the translational degrees of freedom decouple. The problem simplifies to finding the minimum-time trajectory for a 1D drive from an initial state $(x, v) = (0, 0)$ at $t = 0$ to a final state $(x, v) = (L, 0)$ at the final time t_f.

A. The Variational Method and the Hamiltonian

To find the optimal control history—in this case, the acceleration profile $a(t)$—we use the calculus of variations, specifically Pontryagin's Maximum Principle (PMP) [13]. We use the normal maximization convention, in which the Hamiltonian is written as

$$\mathcal{H} = \boldsymbol{\lambda}^\top \mathbf{f}(\mathbf{x}, \mathbf{u}) - L. \tag{3}$$

For a minimum-time problem, the performance index is

$$J = \int_0^{t_f} 1 \, dt,$$

so the running cost is $L = 1$.[2] For this autonomous problem with fixed terminal state and free final time, transversality implies $\mathcal{H} \equiv 0$. A moving terminal state instead requires $\mathcal{H}(t_f) = \lambda_f \cdot \dot{x}_d(t_f)$.

Here:

- L is the **running cost**. For the minimum-time problem considered here, $L = 1$.
- $\mathbf{x}$ is the **state vector**, containing the variables that describe the system.
- $\mathbf{u}$ is the **control vector**, containing the variables we can change to steer the system.
- $\mathbf{f}(\mathbf{x}, \mathbf{u}) = \dot{\mathbf{x}}$ is the vector of **state dynamics**.
- $\boldsymbol{\lambda}$ is the **costate vector**, a set of auxiliary variables introduced by the PMP.

B. Hamiltonian Construction for the Free-Space Case

We apply the principle from Section III-A. First, we identify the components of the 1D problem:

- **State vector:** $\mathbf{x} = (x, v, m)^\top$.
- **Control variable:** $u = a(t)$.
- **State dynamics** $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$:

$$\dot{x} = v, \tag{4}$$

$$\dot{v} = a, \tag{5}$$

$$\dot{m} = -\frac{m^2 a^2}{2P}. \tag{6}$$

- **Costate vector:** $\boldsymbol{\lambda} = (\lambda_x, \lambda_v, \lambda_m)^\top$.

The term $\boldsymbol{\lambda}^\top \dot{\mathbf{x}}$ in the Hamiltonian expands to:

$$\boldsymbol{\lambda}^\top \dot{\mathbf{x}} = \lambda_x \dot{x} + \lambda_v \dot{v} + \lambda_m \dot{m} = \lambda_x v + \lambda_v a - \lambda_m \frac{m^2 a^2}{2P}.$$

[2]Many modern aerospace optimal-control texts write $\mathcal{H} = \boldsymbol{\lambda}^\top \mathbf{f} + L$ and minimize; that convention differs from (3) by the sign of all costates. The present maximization form follows Pontryagin's original formulation [13].

Subtracting the running cost $L = 1$, according to the convention in (3), gives the full Hamiltonian

$$\mathcal{H} = \lambda_x v + \lambda_v a - \lambda_m \frac{m^2 a^2}{2P} - 1. \tag{7}$$

From this Hamiltonian, we derive the costate (adjoint) equations, given by $\dot{\boldsymbol{\lambda}} = -\partial \mathcal{H} / \partial \mathbf{x}$:

$$\dot{\lambda}_x = -\frac{\partial \mathcal{H}}{\partial x} = 0, \tag{8a}$$

$$\dot{\lambda}_v = -\frac{\partial \mathcal{H}}{\partial v} = -\lambda_x, \tag{8b}$$

$$\dot{\lambda}_m = -\frac{\partial \mathcal{H}}{\partial m} = \lambda_m \frac{m a^2}{P}. \tag{8c}$$

C. Optimal Control and First Integrals

According to the PMP, the optimal control $a(t)$ must maximize the Hamiltonian at all times. We find it by setting $\partial \mathcal{H} / \partial a = 0$:

$$\frac{\partial \mathcal{H}}{\partial a} = \lambda_v - \lambda_m \frac{m^2 a}{P} = 0 \implies a(t) = \frac{P \lambda_v(t)}{\lambda_m(t) m^2(t)}. \tag{9}$$

The stationary point is a maximum when $\lambda_m > 0$ (equivalently, $C_m \equiv \lambda_m m^2 > 0$), making the Hamiltonian concave in a. From the costate dynamics, we can extract two integrals of motion. First, (8a) directly implies:

$$\lambda_x(t) = C_1 = \text{const.} \tag{10}$$

Substituting this into (8b) and integrating gives:

$$\lambda_v(t) = -C_1 t + C_2, \tag{11}$$

showing that the velocity costate is a linear function of time.

A second, less obvious, integral is found by examining the time derivative of the product $\lambda_m m^2$:

$$\begin{aligned} \frac{d}{dt}(\lambda_m m^2) &= \dot{\lambda}_m m^2 + \lambda_m (2 m \dot{m}) \\ &= \left(\lambda_m \frac{m a^2}{P}\right) m^2 + \lambda_m \left(2m \left(-\frac{m^2 a^2}{2P}\right)\right) \\ &= \frac{\lambda_m m^3 a^2}{P} - \frac{\lambda_m m^3 a^2}{P} = 0. \end{aligned} \tag{12}$$

Thus, in this uncapped problem, the product $\lambda_m(t) m^2(t)$ is constant throughout the trajectory:

$$\lambda_m(t) m^2(t) = C_m = \text{const.} \tag{13}$$

Substituting these integrals into the control law (9), we discover that the optimal acceleration profile must be linear in time:

$$a(t) = \frac{P}{C_m}(-C_1 t + C_2) = K_1 t + K_0, \tag{14}$$

where $K_1 = -P C_1 / C_m$ and $K_0 = P C_2 / C_m$ are constants.

D. Deriving the Trajectory

We now determine the constants K_1 and K_0 by applying the boundary conditions. Let the final time be t_f. First, we integrate $a(t)$ to find the velocity $v(t)$:

$$v(t) = \int_0^t a(\tau)d\tau + v(0) = \tfrac{1}{2}K_1 t^2 + K_0 t. \tag{15}$$

Applying the final velocity condition $v(t_f) = 0$:

$$\tfrac{1}{2}K_1 t_f^2 + K_0 t_f = 0 \implies K_0 = -\tfrac{1}{2}K_1 t_f. \tag{16}$$

Substituting this back into $a(t)$ reveals its symmetric nature, a characteristic feature of such time-optimal "there-and-back" problems:

$$a(t) = K_1 t - \tfrac{1}{2}K_1 t_f = K_1(t - t_f/2). \tag{17}$$

Next, we integrate $v(t)$ to find the position $x(t)$:

$$x(t) = \int_0^t v(\tau)d\tau + x(0) = \tfrac{1}{6}K_1 t^3 + \tfrac{1}{2}K_0 t^2. \tag{18}$$

Applying the final position condition $x(t_f) = L$:

$$L = \tfrac{1}{6}K_1 t_f^3 + \tfrac{1}{2}K_0 t_f^2. \tag{19}$$

Substituting $K_0 = -\frac{1}{2}K_1 t_f$ into the equation for L allows us to solve for K_1:

$$\begin{aligned} L &= \tfrac{1}{6}K_1 t_f^3 + \tfrac{1}{2}(-\tfrac{1}{2}K_1 t_f)t_f^2 \\ &= (\tfrac{1}{6} - \tfrac{1}{4})K_1 t_f^3 = -\tfrac{1}{12}K_1 t_f^3. \end{aligned} \tag{20}$$

This gives $K_1 = -12L/t_f^3$. Consequently, $K_0 = -\frac{1}{2}(-\frac{12L}{t_f^3})t_f = 6L/t_f^2$. The optimal acceleration is therefore:

$$a(t) = \frac{12L}{t_f^3}\Big(\frac{t_f}{2} - t\Big). \tag{21}$$

This corresponds to an initial acceleration phase, followed by a symmetric braking phase, with $a(t)$ crossing zero at mid-time, $t = t_f/2$. By substituting the constants back and simplifying, we obtain the full trajectory:

$$v(t) = \frac{6Lt}{t_f^2}\Big(1 - \frac{t}{t_f}\Big), \tag{22}$$

$$x(t) = L\Big[3\big(\tfrac{t}{t_f}\big)^2 - 2\big(\tfrac{t}{t_f}\big)^3\Big], \tag{23}$$

which describes a cubic-in-time "S-curve" path.

E. Mass Evolution and Minimum Time

The mass evolution is governed by the separable differential equation (6). Integrating gives:

$$\begin{aligned} \int_{m_i}^{m(t)} \frac{dm}{m^2} &= -\int_0^t \frac{a(\tau)^2}{2P}d\tau \\ \implies \frac{1}{m_i} - \frac{1}{m(t)} &= -\frac{1}{2P}\int_0^t a(\tau)^2 d\tau. \end{aligned} \tag{24}$$

The integral of $a(t)^2$ is:

$$\begin{aligned} \int_0^t a(\tau)^2 d\tau &= \int_0^t \left(\frac{12L}{t_f^3}\right)^2 \left(\frac{t_f}{2} - \tau\right)^2 d\tau \\ &= \frac{144L^2}{t_f^6}\left[-\frac{1}{3}\left(\frac{t_f}{2} - \tau\right)^3\right]_0^t \\ &= \frac{48L^2}{t_f^6}\left(\frac{t_f^3}{8} - \left(\frac{t_f}{2} - t\right)^3\right). \end{aligned} \tag{25}$$

Defining

$$f(s) = 4s^3 - 6s^2 + 3s, \tag{26}$$

the mass relation becomes:

$$\frac{1}{m(t)} - \frac{1}{m_i} = \frac{6L^2 f(t/t_f)}{P\,t_f^3}. \tag{27}$$

The total propellant consumed is found by setting $t = t_f$ (or $s = 1$). If the entire propellant reserve is used, we have $m(t_f) = m_{\rm dry}$, which defines the minimum possible flight-time for the given distance L and mass fraction:

$$\Delta \equiv \frac{1}{m_{\rm dry}} - \frac{1}{m_i} \tag{28}$$

calculated as

$$\Delta = \frac{6L^2 f(1)}{P\,T_{\min}^3} = \frac{6L^2}{P\,T_{\min}^3}. \tag{29}$$

Solving for $T_{\min}$ yields the final result:

$$T_{\min} = \left[\frac{6L^2}{P\Delta}\right]^{1/3}. \tag{30}$$

Equation (30) treats the dry mass as prescribed. If the propulsion hardware mass is instead allowed to scale with power, the same free-space benchmark can be extended by a simple dry-mass closure; the resulting scaling law is derived in Appendix B.

IV. Finite-Exhaust-Speed Extension in Free Space

The uncapped free-space solution of Sec. III requires

$$v_e(t) = \frac{2P}{m(t)|a(t)|}, \tag{31}$$

and therefore demands arbitrarily large exhaust speed when the affine acceleration passes through zero. This section states the resulting finite-ceiling structure needed for interpreting the ideal model. The corrected PMP derivation, switching analysis, closed-form arc maps, and shooting reduction are given in Appendix A.

We impose

$$v_e \le V, \qquad V \equiv v_{e,\max}, \tag{32}$$

under a two-mode engine model: when on, the engine supplies the fixed useful jet power P; when off, the spacecraft coasts. Define

$$F \equiv \frac{2P}{V}, \qquad \sigma \equiv \frac{2P}{V^2}. \tag{33}$$

At full power the ceiling is equivalent to a minimum nonzero acceleration magnitude, so the admissible signed controls are

$$\mathcal{U}(m) = \{0\} \cup \left(-\infty, -\frac{F}{m}\right] \cup \left[\frac{F}{m}, \infty\right). \tag{34}$$

Thus accelerations with magnitude between zero and F/m cannot be produced at fixed power; the engine must instead be switched off.

A. Pointwise Selector and Corrected Adjoint

Set $C = \lambda_m m^2 > 0$ and $p = P\lambda_v/C$, using the current value of C. With $q = F/m$, the selector is

$$a^* = \begin{cases} p, & |p| \geq q, \\ \operatorname{sgn}(p)q, & q/2 < |p| < q, \\ 0, & |p| < q/2. \end{cases} \tag{35}$$

At $|p| = q/2$, coast and the corresponding boundary burn both maximize. The raw velocity costate remains affine, $\lambda_v = n(t_z - t)$, $n = \lambda_x > 0$. The scaled primer p does not: C is constant only on interior burns and coasts. On a maximum-exhaust boundary,

$$\dot{\lambda}_m = \frac{F|\lambda_v|}{m^2}, \qquad \dot{C} = F|\lambda_v| - \frac{2\sigma C}{m}. \tag{36}$$

This follows by differentiating the optimized Hamiltonian with the mass-dependent boundary $a = \pm F/m$ substituted. Holding a fixed while differentiating would omit the motion of the admissible set. For $z = Vm|\lambda_v|/(2C)$, equivalently $\dot{C} = F|\lambda_v|(1 - 1/z)$.

B. Ceiling-Only Structure and Endpoint Reduction

Appendix A proves the ordered pattern

$$[U_+]\ B_+\ C\ B_-\ [U_-\ [B_-]], \tag{37}$$

with absent intervals and endpoint truncations allowed. Here U denotes an interior burn, B a maximum-exhaust burn, and C a coast. The central coast contains the zero of the raw primer. On the braking side the corrected vector fields point strictly upward at $z = 1/2$, so the branch cannot return to this coast threshold, so there is no sliding branch. The proof and a standalone coast/boundary exchange lemma are in the appendix.

Normalize $n = 1$. Shoot in $(t_z, \lambda_m(0))$, propagate state and mass costate with the current selector until $m = m_{\mathrm{dry}}$, and solve $x(t_f) = L$, $v(t_f) = 0$. A positive normal costate scaling then enforces $\mathcal{H} = 0$; competing candidates must still be compared. The fixture $P = 1/2$, $V = 2$, $m_0 = 1$, $m_d = 1/2$, $L = 1$ gives $t_f = 2.35199587$ and is checked by direct transcription in the appendix. This theorem is ceiling-only. A lower exhaust bound introduces maximum-acceleration arcs whose switching topology is left open.

The following Sun-field calculations use the ideal uncapped model. Their implied exhaust speed is

$$v_e(t) = \frac{2P}{m(t)\|\mathbf{a}_T(t)\|}. \tag{38}$$

For a heliocentric finite-ceiling extension, apply the vector selector in the direction of $\boldsymbol{\lambda}_v$ and propagate $\dot{\lambda}_m = F\|\boldsymbol{\lambda}_v\|/m^2$ on boundary burns. The corresponding C varies there. Substituting a vector selector while retaining the ideal constant-C adjoint would be inconsistent.

V. Full Planar Sun-Field Model

We now reinstate solar gravity ($\mu = GM_\odot$) and adopt polar coordinates $\mathbf{x} = (r, \theta)$ in the ecliptic. The control is the *planar thrust acceleration* $\mathbf{a}_T = (a_r, a_\theta)$ produced by the variable-I_{sp} engine. The state vector and its dynamics are:

$$\dot{r} = v_r, \tag{39a}$$
$$\dot{\theta} = \frac{v_\theta}{r}, \tag{39b}$$
$$\dot{v}_r = \frac{v_\theta^2}{r} - \frac{\mu}{r^2} + a_r, \tag{39c}$$
$$\dot{v}_\theta = -\frac{v_r v_\theta}{r} + a_\theta, \tag{39d}$$
$$\dot{m} = -\frac{m^2}{2P}\left(a_r^2 + a_\theta^2\right). \tag{39e}$$

A. Pontryagin Maximum Principle (PMP)

We again apply the variational framework from Section III-A. For this time-optimal problem, the performance index is $J = \displaystyle\int_0^{t_f} \mathrm{d}t$, meaning the Lagrangian is $L = 1$. The components for the Hamiltonian construction are now:

- **State vector:** $\mathbf{x} = (r, \theta, v_r, v_\theta, m)^\top$.
- **Control vector:** $\mathbf{u} = \mathbf{a}_T = (a_r, a_\theta)^\top$.
- **Costate vector:** $\boldsymbol{\lambda} = (\lambda_r, \lambda_\theta, \lambda_{v_r}, \lambda_{v_\theta}, \lambda_m)^\top$.

Using the maximization convention from (3), the Hamiltonian is

$$\mathcal{H} = \boldsymbol{\lambda}^\top \dot{\mathbf{x}} - 1,$$

and it must be maximized by the optimal control at all times. The dot product term $\boldsymbol{\lambda}^\top \dot{\mathbf{x}}$ is constructed by multiplying each costate by its corresponding state derivative from (39):

$$\begin{aligned} \boldsymbol{\lambda}^\top \dot{\mathbf{x}} = \lambda_r v_r + \lambda_\theta \frac{v_\theta}{r} + \lambda_{v_r}\left(\frac{v_\theta^2}{r} - \frac{\mu}{r^2} + a_r\right) \\ + \lambda_{v_\theta}\left(-\frac{v_r v_\theta}{r} + a_\theta\right) - \lambda_m \frac{m^2}{2P}(a_r^2 + a_\theta^2). \end{aligned} \tag{40}$$

Subtracting the running cost $L = 1$ gives the complete Hamiltonian for the Sun-field model. For a minimum-time problem, the PMP asserts that along an optimal trajectory:

1) The state and costates evolve according to Hamilton's canonical equations: $\dot{\mathbf{x}} = \partial\mathcal{H}/\partial\boldsymbol{\lambda}$ and $\dot{\boldsymbol{\lambda}} = -\partial\mathcal{H}/\partial\mathbf{x}$.
2) The optimal control $\mathbf{a}_T(t)$ maximizes $\mathcal{H}$ point-wise.
3) The Hamiltonian remains constant and equal to zero: $\mathcal{H} \equiv 0$.

B. Optimal Control Law

To find the control that maximizes $\mathcal{H}$, we set the partial derivatives of the Hamiltonian with respect to the control variables (a_r, a_θ) to zero:

$$\frac{\partial \mathcal{H}}{\partial a_r} = \lambda_{v_r} - \lambda_m \frac{m^2}{P} a_r = 0, \tag{41a}$$

$$\frac{\partial \mathcal{H}}{\partial a_\theta} = \lambda_{v_\theta} - \lambda_m \frac{m^2}{P} a_\theta = 0. \tag{41b}$$

This yields the optimal thrust acceleration vector:

$$\mathbf{a}_T = k(t)\,\boldsymbol{\lambda}_v, \quad \text{where} \quad \boldsymbol{\lambda}_v \equiv (\lambda_{v_r}, \lambda_{v_\theta})^\top, \tag{42}$$

and the time-varying gain factor is defined as

$$k(t) \equiv \frac{P}{\lambda_m(t) m^2(t)}. \tag{43}$$

The optimal thrust direction is parallel to the velocity costate vector $\boldsymbol{\lambda}_v$. We now show that the gain factor $k(t)$ is, in fact, a constant of motion. Recent adaptive algorithms for low-thrust trajectory design exploit similar primer-vector structures to generate high-quality initial guesses for complex transfers [2].

C. Costate Equations and First Integrals

The costate dynamics are found from $\dot{\lambda}_i = -\partial \mathcal{H}/\partial x_i$:

$$\dot{\lambda}_r = \frac{\lambda_\theta v_\theta}{r^2} + \frac{\lambda_{v_r} v_\theta^2}{r^2} - 2\lambda_{v_r}\frac{\mu}{r^3} - \lambda_{v_\theta}\frac{v_r v_\theta}{r^2}, \tag{44a}$$

$$\dot{\lambda}_\theta = -\frac{\partial \mathcal{H}}{\partial \theta} = 0, \tag{44b}$$

$$\dot{\lambda}_{v_r} = -\lambda_r + \lambda_{v_\theta}\frac{v_\theta}{r}, \tag{44c}$$

$$\dot{\lambda}_{v_\theta} = -\frac{\lambda_\theta}{r} - 2\lambda_{v_r}\frac{v_\theta}{r} + \lambda_{v_\theta}\frac{v_r}{r}, \tag{44d}$$

$$\dot{\lambda}_m = -\frac{\partial \mathcal{H}}{\partial m} = \lambda_m \frac{m}{P}\left(a_r^2 + a_\theta^2\right). \tag{44e}$$

From these dynamics, we extract two first integrals.

a) Rotational Symmetry: Equation (44b) arises because the Hamiltonian is independent of the absolute angle θ (a cyclic coordinate). This gives our first integral:

$$\boxed{\lambda_\theta(t) = C_\theta = \text{const}}. \tag{45}$$

b) Mass–Costate Relationship: We find a direct relationship between mass m and its costate λ_m by examining their rates of change. Using the chain rule, we have $\frac{\mathrm{d}\lambda_m}{\mathrm{d}m} = \frac{\dot{\lambda}_m}{\dot{m}}$. Substituting from (44e) and (39):

$$\frac{\mathrm{d}\lambda_m}{\mathrm{d}m} = \frac{\lambda_m \frac{m}{P}\left(a_r^2 + a_\theta^2\right)}{-\frac{m^2}{2P}\left(a_r^2 + a_\theta^2\right)} = \frac{\lambda_m m}{-m^2/2} = -2\frac{\lambda_m}{m}.$$

This is a separable first-order ordinary differential equation:

$$\frac{\mathrm{d}\lambda_m}{\lambda_m} = -2\frac{\mathrm{d}m}{m}.$$

Integrating both sides yields $\ln(\lambda_m) = -2\ln(m) + \text{const}$, which upon exponentiation gives our second, crucial integral of motion:

$$\boxed{\lambda_m(t) m^2(t) = C = \text{const}}. \tag{46}$$

This relation holds for the entire trajectory.

c) Constant Thrust Gain: Substituting the integral from (46) back into the definition of the gain factor (43), we find that it is constant:

$$k(t) = \frac{P}{\lambda_m(t) m^2(t)} = \frac{P}{C} \equiv k_T = \text{const}. \tag{47}$$

The optimal control law (42) therefore simplifies to:

$$\mathbf{a}_T(t) = k_T\,\boldsymbol{\lambda}_v(t). \tag{48}$$

The physically relevant normal branch has $C \equiv \lambda_m m^2 > 0$, so $k_T = P/C > 0$ and the Hamiltonian is concave in the thrust-acceleration components, confirming that the stationary control maximizes $\mathcal{H}$. The thrust magnitude is proportional to $\|\boldsymbol{\lambda}_v\|$, and the constant gain k_T is determined by the boundary conditions—a key result that significantly simplifies the nature of the optimal control.

This completes the analytical formulation. After using the two first integrals, the orbital and costate shooting dynamics are governed by seven differential equations: the four kinematic/dynamic variables $(r, \theta, v_r, v_\theta)$ and the three non-constant costates $(\lambda_r, \lambda_{v_r}, \lambda_{v_\theta})$. The mass need not be integrated as an independent shooting state, because it can be recovered from the accumulated inverse-mass expenditure,

$$\frac{1}{m(t)} - \frac{1}{m_i} = \int_0^t \frac{a_r(\tau)^2 + a_\theta(\tau)^2}{2P}\, d\tau. \tag{49}$$

For a dry-mass terminal extremal the required scalar quadrature equality is

$$\int_0^{t_f} \frac{a_r(t)^2 + a_\theta(t)^2}{2P}\, dt = \Delta, \qquad \Delta = \frac{1}{m_{\text{dry}}} - \frac{1}{m_i}. \tag{50}$$

Section VII-C separates this required equality from the historical penalty implementations and verifies it for the reconstructed candidates. A shooting method must find the initial values of the costates, the constant C_θ, the thrust-gain constant k_T, and the final time t_f so that the final circular-orbit boundary conditions and the mass-budget constraint are satisfied.

VI. Similarity Reduction of the Sun-Field Transfer Family

The Sun-field boundary-value problem from Sec. V is written in dimensional variables, but its family of circular-to-circular transfers possesses a useful scaling symmetry. This symmetry allows the six nominal inputs

$$(r_0,\ r_f,\ \Theta,\ P,\ m_i,\ m_{\text{dry}})$$

to be reduced to three dimensionless parameters for the ideal uncapped model. If a finite exhaust-speed ceiling $v_{e,\max}$ is imposed, two additional dimensionless parameters enter: the normalized ceiling $\nu_{\max} \equiv v_{e,\max}/\sqrt{\mu/r_0}$ and the initial-to-dry mass ratio $\eta \equiv m_i/m_{\text{dry}}$, because the ceiling condition $v_e = 2P/(m\|\mathbf{a}_T\|)$ mixes mass with the other propulsion quantities in a way that κ alone cannot absorb. The capped model therefore depends on $(\rho, \kappa, \Theta, \nu_{\max}, \eta)$ rather than (ρ, κ, Θ).

The reduction is useful both conceptually and numerically: conceptually, because it shows which combinations of the

physical inputs actually determine the transfer shape; numerically, because it makes it possible to tabulate a reduced atlas of normalized solutions and then map them back to dimensional trajectories for any specific spacecraft.

A. Canonical units and normalized variables

Let the departure radius define the distance unit,

$$R_0 \equiv r_0, \tag{51}$$

and let the associated Keplerian time and velocity scales be

$$T_0 \equiv \sqrt{\frac{R_0^3}{\mu}}, \qquad V_0 \equiv \frac{R_0}{T_0} = \sqrt{\frac{\mu}{R_0}}. \tag{52}$$

The corresponding acceleration scale is

$$A_0 \equiv \frac{V_0}{T_0} = \frac{\mu}{R_0^2}. \tag{53}$$

We now define the dimensionless independent variable and normalized state by

$$\tau \equiv \frac{t}{T_0}, \qquad \bar{r} \equiv \frac{r}{R_0}, \qquad \bar{v}_r \equiv \frac{v_r}{V_0}, \qquad \bar{v}_\theta \equiv \frac{v_\theta}{V_0}, \tag{54}$$

and similarly normalize the thrust acceleration components as

$$\bar{a}_r \equiv \frac{a_r}{A_0}, \qquad \bar{a}_\theta \equiv \frac{a_\theta}{A_0}. \tag{55}$$

The polar angle θ is already dimensionless and is left unchanged.

The mass is handled more conveniently through the inverse-mass increment that already appears in the free-space solution. Define

$$\Delta \equiv \frac{1}{m_{\mathrm{dry}}} - \frac{1}{m_i}, \tag{56}$$

and introduce the normalized fuel coordinate

$$q(\tau) \equiv \frac{m(\tau)^{-1} - m_i^{-1}}{\Delta}. \tag{57}$$

By construction,

$$q(0) = 0, \qquad q(\tau_f) = 1, \tag{58}$$

so q measures the fraction of the available inverse-mass budget that has been consumed. The dimensional mass is recovered from q as

$$m(\tau) = \frac{1}{m_i^{-1} + \Delta q(\tau)}. \tag{59}$$

B. Dimensionless form of the dynamics

Substituting the scaled variables into (39) gives

$$\bar{r}' = \bar{v}_r, \tag{60a}$$

$$\theta' = \frac{\bar{v}_\theta}{\bar{r}}, \tag{60b}$$

$$\bar{v}_r' = \frac{\bar{v}_\theta^2}{\bar{r}} - \frac{1}{\bar{r}^2} + \bar{a}_r, \tag{60c}$$

$$\bar{v}_\theta' = -\frac{\bar{v}_r \bar{v}_\theta}{\bar{r}} + \bar{a}_\theta, \tag{60d}$$

$$q' = \frac{\bar{a}_r^2 + \bar{a}_\theta^2}{\kappa}, \tag{60e}$$

where prime denotes differentiation with respect to τ, and where the single dimensionless propulsion parameter is

$$\kappa \equiv 2P\Delta \frac{R_0^{5/2}}{\mu^{3/2}} = 2P \left(\frac{1}{m_{\mathrm{dry}}} - \frac{1}{m_i} \right) \frac{r_0^{5/2}}{\mu^{3/2}}. \tag{61}$$

This is the key reduction. In dimensional form, the mass-flow law (2) depends separately on P, m_i, m_{dry}, and r_0. After scaling, those four quantities appear only through the single combination κ. The normalized equations therefore no longer distinguish between spacecraft that have different powers and masses but the same value of κ. Such spacecraft follow the same dimensionless trajectory and differ only by the dimensional scales used to reconstruct time, radius, velocity, and acceleration.

C. Dimensionless boundary conditions

For a transfer from a circular orbit of radius r_0 to a circular orbit of radius r_f, the normalized boundary conditions become

$$\bar{r}(0) = 1, \qquad \bar{r}(\tau_f) = \rho, \tag{62a}$$

$$\theta(0) = 0, \qquad \theta(\tau_f) = \Theta, \tag{62b}$$

$$\bar{v}_r(0) = 0, \qquad \bar{v}_r(\tau_f) = 0, \tag{62c}$$

$$\bar{v}_\theta(0) = 1, \qquad \bar{v}_\theta(\tau_f) = \rho^{-1/2}, \tag{62d}$$

$$q(0) = 0, \qquad q(\tau_f) = 1, \tag{62e}$$

with

$$\rho \equiv \frac{r_f}{r_0}. \tag{63}$$

The quantity ρ is the radius ratio, while

$$\Theta \equiv \theta_f - \theta_0 + 2\pi N \tag{64}$$

is the unwrapped transfer angle, with $N \in \mathbb{Z}_{\geq 0}$ selecting the number of completed revolutions. Only the angular difference matters, not the absolute departure angle, because the Sun-field model is rotationally invariant. This is the same symmetry that produced the first integral $\lambda_\theta = C_\theta$ in Sec. V.

Equations (60)–(62) show that, for the uncapped variable-I_{sp} model, the normalized transfer problem depends only on

$$(\rho,\ \kappa,\ \Theta).$$

This is the reduced parameter space for the ideal uncapped model.

D. Why the parameter space shrinks

It is useful to state explicitly what has been removed.

First, the absolute orbital scale r_0 disappears from the normalized equations because lengths are measured in units of r_0, times in units of $\sqrt{r_0^3/\mu}$, and velocities in units of $\sqrt{\mu/r_0}$. Only the ratio r_f/r_0 remains.

Second, the three propulsion quantities P, m_i, and m_{dry} do not independently affect the normalized dynamics. They enter only through the inverse-mass budget Δ and then only through the combination κ in (61). Thus three nominal propulsion parameters collapse to one similarity parameter.

Third, the absolute phase angle θ_0 is irrelevant because the equations are invariant under a rigid rotation of the entire trajectory. Only the total angular change Θ matters.

As a result, the circular-to-circular Sun-field transfer problem is parameterized, at the similarity level, by the three variables (ρ, κ, Θ) rather than the six nominal quantities $(r_0, r_f, \Theta, P, m_i, m_{\rm dry})$.

E. Forward map from nominal parameters to similarity space

For a given dimensional transfer request, the reduced coordinates are obtained directly from the nominal mission parameters:

$$\rho = \frac{r_f}{r_0}, \tag{65}$$

$$\Theta = \theta_f - \theta_0 + 2\pi N, \tag{66}$$

$$\Delta = \frac{1}{m_{\rm dry}} - \frac{1}{m_i}, \tag{67}$$

$$\kappa = 2P\Delta\,\frac{r_0^{5/2}}{\mu^{3/2}}. \tag{68}$$

Here N selects the desired multi-revolution branch. Thus the similarity reduction is applied by first computing (ρ, κ, Θ) from the physical problem and then solving the corresponding normalized boundary-value problem.

F. Inverse map from a normalized atlas solution to dimensional variables

Suppose now that a normalized solution has been computed or interpolated in the reduced space (ρ, κ, Θ). That solution consists of the dimensionless functions

$$\bar r(\tau), \quad \theta(\tau), \quad \bar v_r(\tau), \quad \bar v_\theta(\tau), \quad q(\tau), \quad \bar a_r(\tau), \quad \bar a_\theta(\tau),$$

together with the normalized final time τ_f.

For the specific dimensional system defined by r_0, m_i, $m_{\rm dry}$, and μ, the trajectory is recovered by the inverse scaling

$$t = T_0\tau = \sqrt{\frac{r_0^3}{\mu}}\,\tau, \tag{69}$$

$$r(t) = r_0\,\bar r(\tau), \tag{70}$$

$$\theta(t) = \theta_0 + \theta(\tau), \tag{71}$$

$$v_r(t) = \sqrt{\frac{\mu}{r_0}}\,\bar v_r(\tau), \tag{72}$$

$$v_\theta(t) = \sqrt{\frac{\mu}{r_0}}\,\bar v_\theta(\tau), \tag{73}$$

$$a_r(t) = \frac{\mu}{r_0^2}\,\bar a_r(\tau), \tag{74}$$

$$a_\theta(t) = \frac{\mu}{r_0^2}\,\bar a_\theta(\tau), \tag{75}$$

$$m(t) = \frac{1}{m_i^{-1} + \Delta q(\tau)}. \tag{76}$$

In particular, the dimensional flight time is

$$t_f = \sqrt{\frac{r_0^3}{\mu}}\,\tau_f. \tag{77}$$

This gives the practical recipe for using a precomputed solution atlas:

1) Start from the nominal physical parameters.
2) Compute ρ, Θ, and κ from the forward map above.
3) Retrieve or interpolate the normalized solution stored at that location in (ρ, κ, Θ)-space.
4) Recover the dimensional trajectory by the inverse scaling formulas.

G. What should be stored in the atlas

For numerical continuation, it is not necessary to store the full dimensional trajectory. A more efficient choice is to store the normalized shooting data, namely the initial costates, the constant C_θ associated with rotational symmetry, the thrust-gain constant k_T, and the normalized flight time. For the uncapped model, because these quantities are stored in dimensionless form, the same atlas entry can be reused for any dimensional spacecraft that maps to the same (ρ, κ, Θ). If a finite exhaust-speed ceiling is included, the atlas must also be indexed by $\nu_{\max}$ and η. This is the computational advantage of the similarity reduction: one normalized solution represents an entire equivalence class of physical missions.

VII. Numerical Procedure

The historical two-stage procedure uses global search and local continuation to find endpoint-targeting seeds. Its penalty objectives are described first; the corrected equality and normal-transversality acceptance tests follow in Section VII-C [3], [11].

A. Objective Function

The core of the numerical method is an objective function, J, which quantifies endpoint targeting accuracy and constraint satisfaction for a candidate trajectory. This function is minimized to find a set of initial costate parameters that satisfies the endpoint and constraint residuals as accurately as possible. It incorporates penalties for violating physical constraints and errors for missing the target state. The function has two distinct forms depending on the optimization stage.

1) Global Search Objective: In the initial search phase, the goal is to find *any* feasible trajectory whose terminal state lies on some circular heliocentric orbit beyond a minimum radius, without fixing the specific final radius or polar angle. The target radius and final polar angle are pinned only in Stage 2. The Stage 1 objective function J_{global} therefore rewards reaching the local circular velocity at whatever final radius the trajectory attains in the outer solar system, while penalizing physically unrealistic outcomes:

$$J_{\text{global}} = w_f\mathcal{P}_f + w_r\mathcal{P}_r + w_v\mathcal{E}_v^2. \tag{78}$$

The terms are defined as:

- $\mathcal{P}_f$: A fuel penalty that applies a large constant value if the final mass m_f drops below the dry mass m_{dry}, as a heuristic deterrent to overuse; this finite penalty is

neither a hard constraint nor an equality enforcing dry-mass arrival.
- $\mathcal{P}_r$: A radial position penalty, $\max(0, r_{\min} - r_f)^2$, which discourages solutions that fail to travel a minimum distance $r_{\min}$ (e.g., 1.5 AU) from the Sun.
- $\mathcal{E}_v$: The velocity mismatch error at the final state. It measures the Euclidean distance between the final velocity vector $\mathbf{v}_f = (v_{r,f}, v_{\theta,f})$ and the required velocity for a circular orbit at the final radius r_f, given by $\mathbf{v}_{\text{circ}} = (0, \sqrt{\mu/r_f})$. The squared error is $\mathcal{E}_v^2 = v_{r,f}^2 + (v_{\theta,f} - \sqrt{\mu/r_f})^2$.
- w_f, w_r, w_v: Positive weighting constants that balance the contribution of each term.

2) Local Refinement Objective: Once an initial solution is found, the objective function is switched to a targeted version, J_{local}, to steer the trajectory to a precise destination $(r_{\text{des}}, \theta_{\text{des}})$:

$$J_{\text{local}} = w_f \mathcal{P}_f + w_v \mathcal{E}_v^2 + w_r (r_f - r_{\text{des}})^2 + w_\theta (\theta_f - \theta_{\text{des}})^2. \quad (79)$$

Here, the radial penalty $\mathcal{P}_r$ is removed, and two new terms are added to penalize deviations from the desired final radius r_{des} and polar angle θ_{des}.

B. Two-Stage Optimization Strategy

1) Stage 1: Global Search: An initial approximate seed is sought by minimizing the global objective function (78). For this task, we use the `differential_evolution` algorithm from the SciPy library [14]. This robust, population-based method is well-suited for exploring the wide parameter space of the initial costates without getting trapped in local minima, thereby finding a good starting point for the next stage.

2) Stage 2: Continuation and Local Refinement: Starting from the solution found in Stage 1, which ends at a point $(r^\star, \theta^\star)$, we use a continuation (or homotopy) method to incrementally approach the final desired target $(r_{\text{des}}, \theta_{\text{des}})$ [15]. The procedure is as follows:

1) Set an intermediate target partway between the current known solution's endpoint and the final desired endpoint.
2) Perform a local optimization by minimizing the targeted objective function (79) using Powell's method [16] (derivative-free) as implemented in `scipy.optimize.minimize`. The initial guess for the optimizer is the set of parameters from the previous successfully converged solution.
3) The historical implementation accepts a high-precision endpoint fit. In the corrected workflow it must also pass the equality and normal-transversality tests in Section VII-C. The intermediate target is then advanced closer to the final destination, and the process repeats.
4) If the optimizer fails to converge, the cause is unresolved; reducing the continuation step is a recovery heuristic. The intermediate target is then moved back, typically halving the distance between the last successful endpoint and the current target, and the optimization is re-attempted.

This adaptive stepping keeps the local optimizer close to a previously converged solution and substantially improves robustness, although convergence is not guaranteed for every target or every continuation step.

C. Endpoint Equalities and Free-Time Normalization

The two penalty objectives are seed-finding devices, not complete PMP acceptance tests. In particular, a one-sided dry-mass penalty vanishes with unused propellant and neither objective explicitly minimizes time. The pinned implementation has two different paths: its event-driven integrator stops at $m = m_d$ (unless another event stops it first), whereas its fixed-time boundary residual uses only $\max(0, m_d - m_f)$. The latter does not enforce dry-mass equality, so convergence of that path alone does not give a minimum-time PMP extremal.

For a fixed final polar angle, the corrected reconstruction integrates normalized fixed-time effort costates p, with $u = p_v/2$, and the quadrature $I' = |u|^2$. Shoot in $(p_{r0}, p_{v_r 0}, p_{v_\theta 0}, p_\theta, \log h)$ against the five *equality* residuals

$$\begin{pmatrix} r(h) - \rho \\ \theta(h) - \Theta \\ v_r(h) \\ v_\theta(h) - \rho^{-1/2} \\ I(h) - \kappa \end{pmatrix} = 0. \quad (80)$$

This enforces both circular endpoint state and $m_f = m_d$; h is an unknown. The autonomous effort Hamiltonian is

$$E = p_r v_r + p_\theta v_\theta / r + p_{v_r}(v_\theta^2/r - r^{-2}) - p_{v_\theta} v_r v_\theta / r + |p_v|^2/4. \quad (81)$$

It is constant and initially $E = p_\theta + |p_v(0)|^2/4$. If $E > 0$, the time covectors are p/E and the quadrature covector is $-1/E$. Their normal time Hamiltonian is $\mathcal{H}_T = E/E - 1 = 0$. Thus free-time transversality is an explicit positive-normalization check, not an extra independent equation after removal of costate scale. If $E \le 0$, this construction does not establish a normal minimum-time extremal. For a moving circular target, use $\theta(h) = \phi + \rho^{-3/2} h$ instead and normalize by $E - p_\theta \rho^{-3/2}$; the transversality condition is then $\mathcal{H}_T - \lambda_f \cdot \dot{x}_d = 0$, not generally $\mathcal{H}_T = 0$.

For the 73 independently reconstructed 5.2 AU final-angle rows, E ranges from 4.03123 to 8.78006 and is positive throughout. Maximum normalized endpoint error is $4.8\,10^{-11}$, effort-equality error $1.47\,10^{-10}$, final-mass error $1.3\,10^{-5}$ kg, and maximum reconstructed $|\mathcal{H}_T|$ is $1.1\,10^{-9}$. These rows are therefore normal extremals to numerical precision; their local or global optimality is not checked. The independent figure reconstructions have mass errors below 0.002 kg; the extreme solar pass has maximum Hamiltonian error $2.2\,10^{-6}$.

D. Atlas Sensitivities and Resource Parameters

For a regular ideal minimum-effort branch $F(d, h)$ whose initial and terminal states follow ballistic Kepler ephemerides, endpoint variation gives

$$F_h = -|u_f|^2, \qquad F_d = |u_0|^2 - |u_f|^2. \quad (82)$$

With a fixed final polar angle instead, $F_h = -E$. These are different boundary conditions. The attainable budget-to-time

value is $h^*(\kappa) = \inf\{h : F^*(d,h) \le \kappa\}$ when the threshold is attained; otherwise actual control feasibility must be retained in the definition. A regular moving-endpoint root permits the safeguarded Newton step

$$h_{\text{new}} = h + \frac{F(d,h) - \kappa}{|u_f|^2}. \tag{83}$$

Use a bracket and compare branches; a stationary root is not a global value certificate. These sensitivities are developed in the unified minimum-effort manuscript [17].

For finite bounds let $V_0 = \sqrt{\mu/r_0}$, $J_0 = V_0^2/T_0$, $\chi = 2P/(m_0 J_0)$ and $\eta = m_0/m_d$. Then

$$\begin{aligned} \kappa &= \chi(\eta - 1), \qquad \nu_\pm = v_{e,\pm}/V_0, \\ w &= \chi + I, \qquad w/\nu_+ \le |u| \le w/\nu_-. \end{aligned} \tag{84}$$

Coast is separately admissible. The ceiling parameter is $\nu_{\max} = \nu_+$; the ideal atlas parameter κ alone no longer specifies the capped engine. The mass ratio supplies $\chi = \kappa/(\eta - 1)$, and both exhaust bounds must also be retained.

The original solver and plots are available in the repository [18]. The code audit is pinned to commit bdc9698212c92cd0815d9251c7d777d47cefd706. The corrected reconstruction and per-cell audit ledger are distributed with this source. Historical solver acceptance flags are not used as evidence of feasibility or optimality.

VIII. Results

The numerical procedure was applied to compute ideal transfer candidates from a 1 AU circular orbit (representing Earth) to various target circular orbits. The targets are defined by their final radius and the heliocentric polar angle $\theta_f - \theta_0$ relative to the starting position. (We retain the symbol θ for this polar angle throughout; it is not the orbital true anomaly, which is undefined on a circular orbit.) The propulsion model used $P = 1\,\text{GW}$ of useful jet power, with an initial vehicle mass of $3000\,\text{t}$ and a dry mass of $1000\,\text{t}$. These calculations use the ideal uncapped control law of Sec. V; no finite upper bound on v_e is imposed. As an order-of-magnitude check on these numbers, the free-space benchmark of Sec. III with $L = r_f - r_0 = 4.2\,\text{AU}$, $P = 1\,\text{GW}$, and $\Delta = 1/m_{\text{dry}} - 1/m_i$ gives

$$T_{\min} = \left[\frac{6L^2}{P\Delta}\right]^{1/3} \approx 177\ \text{days}.$$

The Sun-field transfers to $5.2\,\text{AU}$ in Table I range from about 170 to 220 days and bracket this value. The historical departure-phase table spans a wider range, extending to about 281 days for the 5.2 AU target. The flight times shorter than the free-space benchmark are possible because the spacecraft begins with a non-zero tangential velocity ($\sim 30\,\text{km/s}$ at 1 AU) and can convert solar gravitational potential energy along the transfer. The longer ones reflect the additional arc length traversed around the Sun, the energy required to climb the potential well to the outer orbit, and, in some cases, the cost of the low-perihelion Oberth pass.

Figures 1 and 2 show the relationship between flight time and the orbital geometry. The flight times exhibit a U-shaped dependence on the final polar angle, with a clear minimum for transfers that are neither direct short-arcs nor long-arcs. Similar behavior has been reported in recent multi-objective studies of variable-I_{sp} maneuvers [11].

Figures 3 and 5 show example candidate extremals. A key feature of these trajectories is the inclusion of an Oberth-like low-perihelion maneuver. To gain orbital energy efficiently, the spacecraft first thrusts to lower its periapsis, making a low-perihelion pass around the Sun. It then applies its main acceleration burn at maximum velocity near the Sun, which is far more effective at increasing orbital energy than thrusting in a higher orbit [19]. This is visible in the plots as an initial dip towards the Sun and a corresponding peak in the acceleration magnitude. These Oberth-type solutions should be interpreted within the idealized dynamical model used here. No explicit constraints are imposed on minimum perihelion distance, solar heating, radiator performance near the Sun, pointing restrictions, or power-system survivability. Adding such constraints would generally increase the transfer times or remove some of the most aggressive solar-pass solutions.

Tables I and II retain the historical outward time grid. Cells marked "U" are unresolved solver outputs, not proven infeasible transfers. Excluding a geometric cell would need a global effort lower bound $L(h) > \kappa$ for every admissible time and winding; solver failure supplies no such bound. Tables III and IV give the historical return-to-1-AU results, which were not re-audited.

A. Continuous-Path Diagnostics and Audit Coverage

The available source tables give rounded flight times but do not identify per-cell initial costates or full trajectories. A flight time and an endpoint alone do not determine a unique path or its exhaust extrema. We therefore distinguish independently reconstructed candidates from the historical plotted and tabulated outputs. The accompanying `verification/table_diagnostics.csv` has one record for every cell, with explicit unavailable and unresolved statuses; blank extrema are not zero. The complete 73-row 5.2 AU final-angle reconstruction includes minimum radius and both exhaust extrema for each row. The other historical cells do not have individually verified path diagnostics. Figures 1–2 are aggregate time plots: only their corresponding reconstructed subset is audited.

The indicated geometries of the three path illustrations give:

ρ, Θ	days	$r_{\min}$ AU	$v_{e,\min}$	$v_{e,\max}$
$1.5, 34.7^\circ$	42.546	1.0000	20.37	993.91
$5, 273.1^\circ$	206.560	0.169926	42.35	13001.03
$5.2, 385.2^\circ$	214.236	0.0002732	4.85	1203.49

Speeds are in km/s. These reconstruct the stated geometries. The first gives 42.5 days against the 44 days of the figure label, so that plotted path is not reproduced exactly.

Radius minima are found from radial-velocity roots. Exhaust extrema are refined from dense samples, including endpoint values. Every band violation is large, so sampling error does not change the verdicts. All 73 Jupiter rows fail the 50–250 km/s band, and rows 305°–360° also cross 0.1 AU. The three indicated figure geometries likewise fail the

band. The extreme pass reaches approximately 0.00027 AU, well outside a physical solar mission model. These are ideal-engine benchmarks. Enforcing an exhaust band or perihelion floor requires a new constrained solve and may change the branch and time; clipping controls or rejecting points after the solve does not repair the endpoint.

A hard radial floor is correctly posed as a state constraint. For $c = (r_{\min}^2 - |x|^2)/2 \le 0$, the maximizing convention can use a nonnegative measure dM supported on contact, with $dp_x = -Dg^T p_v\, dt - x\, dM$, $dp_v = -p_x\, dt$ and the associated entry, exit and tangency conditions [20]. The finite-bound resource adjoint must still be propagated. The companion minimum-effort article specializes these conditions [17].

IX. Conclusion

We combined analytic insight (two closed-form first integrals, an uncapped free-space benchmark, and a finite-exhaust-speed structural extension) with a continuation strategy for constructing time-optimal transfer candidates for a constant-power, variable-I_{sp} engine. The reconstructed subset satisfies normal PMP necessary conditions; the remaining historical numerical entries are retained as unverified solver outputs. The uncapped numerical method generates candidate trajectories, including those featuring low-perihelion Oberth-like maneuvers, for transfers out to the giant planets with a gigawatt-class useful jet-power engine. Recent developments in variable-I_{sp} propulsion and adaptive low-thrust guidance algorithms [2], [11] suggest that further improvements in mission performance are possible. The companion pursuit article treats ideal free-space rendezvous in dimensions two and three and gives local planar CW and nonlinear Kepler capture certificates [21]. A full constrained three-dimensional interplanetary optimizer remains future work. A rigorous demonstration of local or global time optimality for the Sun-field examples remains future work; it would require either second-order sufficiency analysis of the extremals, such as conjugate-point checks, or a broader global comparison across competing transfer branches.

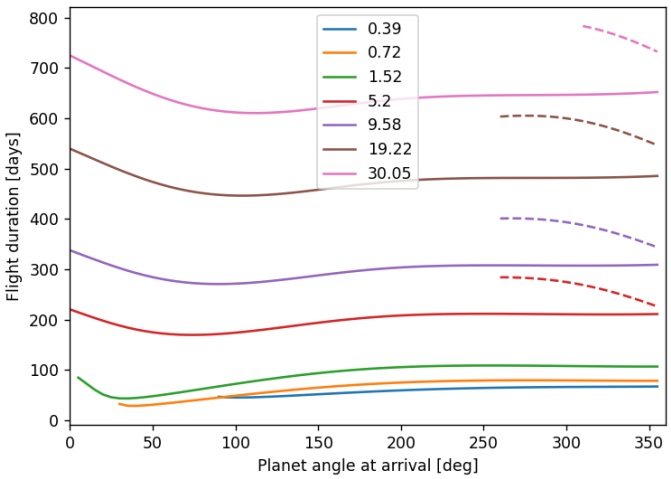


Fig. 1. Flight times in days versus final polar angle at arrival.

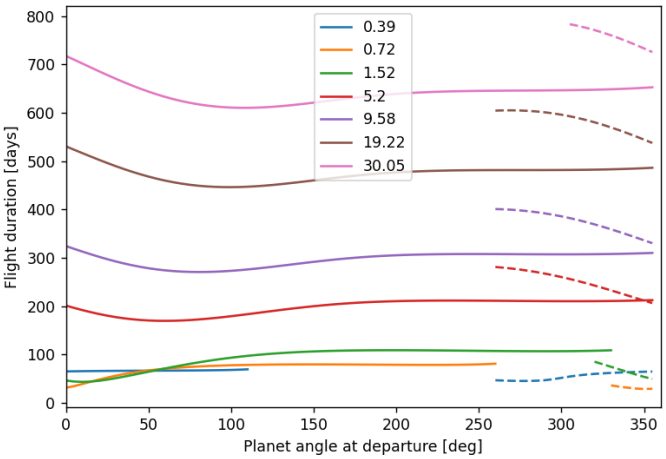


Fig. 2. Flight times in days versus initial polar angle at departure.

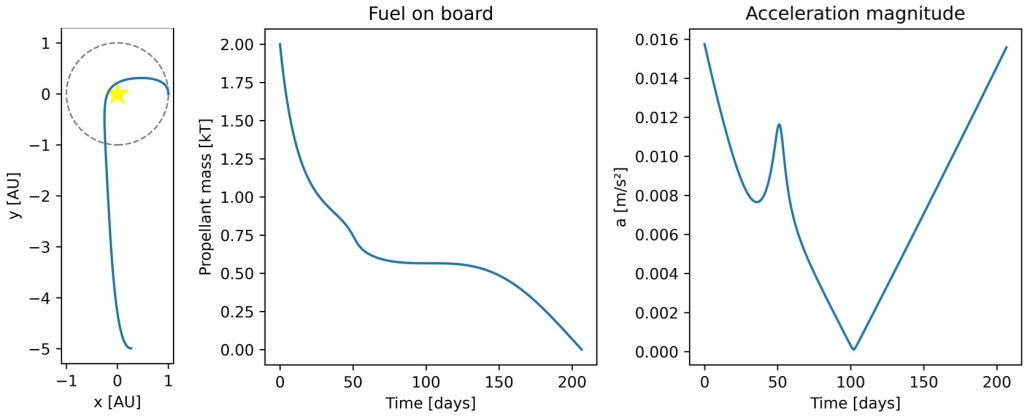


Fig. 3. Example trajectory from $r = 1\,\text{AU}$, $\theta = 0^\circ$ to $r = 5\,\text{AU}$, $\theta = 273.1^\circ$ in 207 days. The trajectory includes a low-perihelion Oberth-like pass to gain orbital energy, which is apparent from the acceleration peak around day 50.

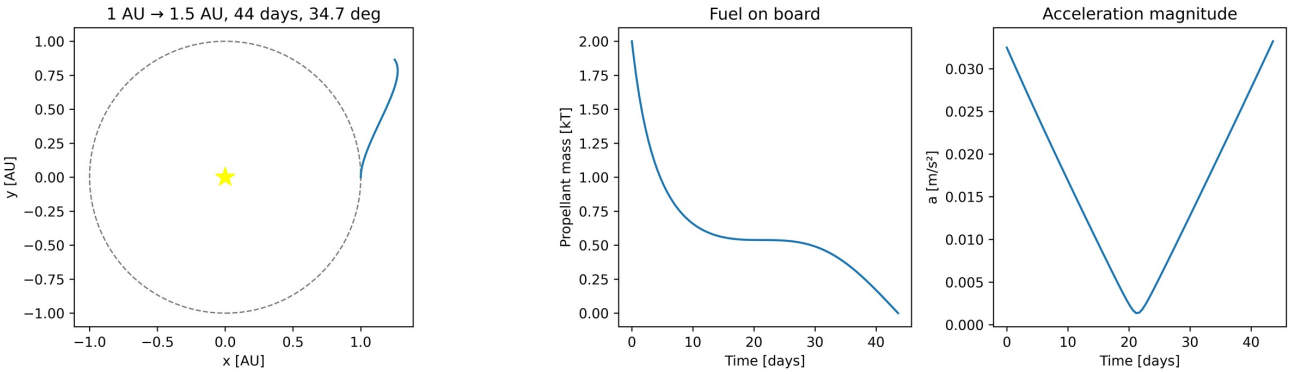


Fig. 4. Example trajectory from $r = 1\,\text{AU}$, $\theta = 0^\circ$ to $r = 1.5\,\text{AU}$, $\theta = 34.7^\circ$ in 44 days. Recomputing the stated rounded geometry gives 42.55 days (Sec. VIII-A), so the plotted path is not reproduced exactly.

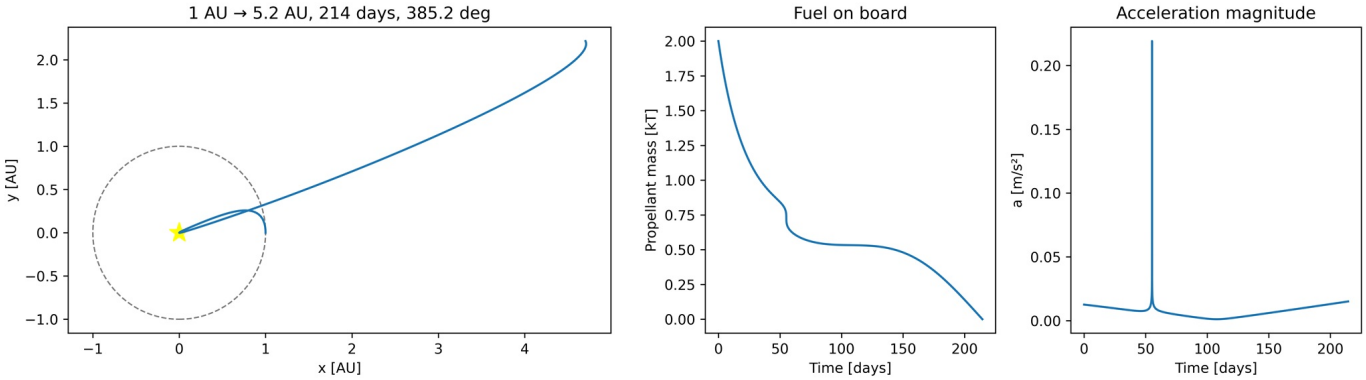


Fig. 5. An example of a more extreme low-perihelion Oberth-like maneuver for a transfer to Jupiter's orbit. The trajectory is from $r = 1\,\text{AU}$, $\theta = 0^\circ$ to $r = 5.2\,\text{AU}$, $\theta = 385.2^\circ$ and lasts 214 days.

ACKNOWLEDGMENT

The author wishes to acknowledge the use of large language models (Google's Gemini and OpenAI's GPT) for assistance in improving the language and clarity of the manuscript, for aiding in the implementation of the numerical code and derivations of some of the analytical results.

APPENDIX A
CORRECTED FINITE-CEILING FREE-SPACE DERIVATION

For the dimensional time problem in one dimension, let $l = \lambda_v$ with $l' = -\lambda_x$ and λ_x constant. On a powered fixed-sign interval, write $|l| = L_a + nt$. For fixed exhaust V set $F = 2P/V$, $\sigma = 2P/V^2$ and $C_m = \lambda_m m^2$. Correct differentiation gives

$$m' = -\sigma, \quad \lambda_m' = \frac{F|l|}{m^2}, \quad C_m' = \frac{2P|l|}{V} - \frac{4PC_m}{mV^2}. \tag{85}$$

Thus C_m is not constant on this boundary. With $B = n/\sigma$ and $A = L_a + Bm_a$,

$$\begin{aligned} m &= m_a - \sigma t, \\ \lambda_m &= \lambda_{m,a} + V\left[A\left(\frac{1}{m} - \frac{1}{m_a}\right) + B\log\frac{m}{m_a}\right]. \end{aligned} \tag{86}$$

For thrust sign $\epsilon = \pm 1$,

$$\begin{aligned} v &= v_a + \epsilon V \log(m_a/m), \\ x &= x_a + v_a t + \epsilon V\left[t - \frac{m}{\sigma}\log(m_a/m)\right]. \end{aligned} \tag{87}$$

On an ideal interior arc C_m is constant and $a = Pl/C_m$; on a coast both m and λ_m are constant.

A. What survives and what is replaced

The pointwise selector survives when evaluated with the current C_m. The fixed-exhaust state maps above, the interior-arc mass formula, and the exchange argument excluding an avoidable powered–coast rearrangement under its monotone braking hypotheses also survive. On an interior arc $l = l_a - nt$,

$$\frac{1}{m(t)} = \frac{1}{m_a} + \frac{P}{2C_m^2}(l_a^2 t - l_a n t^2 + n^2 t^3/3). \tag{88}$$

Integrating $a = Pl/C_m$ gives quadratic velocity and cubic position. The raw primer remains affine globally; its scaled acceleration does not. Closed-form switching roots that assume constant C_m on boundary arcs do not carry over, and are not repaired by substituting a terminal value of C_m.

Lemma 1 (Coast/boundary exchange)**.** *In a monotone nonnegative-velocity braking phase, any positive coast after a positive fixed-exhaust brake can be moved ahead of the burns and shortened to preserve distance, terminal velocity and mass while reducing time.*

Proof. In a coast/ceiling-brake mixture with duty fraction $\delta \in [0,1]$, write $s(t) = \int \delta\, dt$. Then $m = m_a - \sigma s$ and $v = v_a - V\log(m_a/m)$. Final mass and velocity depend only on total powered time. Exchanging a burn followed by a coast of length c for coast followed by burn increases distance by $c\,\Delta v > 0$, where Δv is the burn's velocity decrement. Repetition puts all coasting first. If the braking interval has $0 \le v \le v_a$ and $v_a > 0$, its original coast distance is reproduced by a shorter leading coast of duration $\int_{\text{coast}} v\, dt/v_a$. Mass, terminal velocity and distance are unchanged, while elapsed time strictly decreases whenever a positive coast follows a positive brake. Approximation extends this state argument to fractional mixtures. The argument uses only the state equations. □

B. Ceiling-only topology and the unresolved two-sided case

Consider the ceiling-only, positive-distance, rest-to-rest problem with $C_m > 0$ and positive dry mass. Normalize the constant $\lambda_x = n > 0$; the primer is $l = n(t_z - t)$. Define $z = Vm|l|/(2C_m)$. The selector gives interior U for $z > 1$, ceiling burn B for $1/2 < z < 1$, and coast C for $z < 1/2$. On a ceiling arc,

$$\frac{z'}{z} = \frac{|l|'}{|l|} + \frac{\sigma}{m}(1 - 2z), \qquad \sigma = 2P/V^2. \tag{89}$$

On an interior arc $z'/z = |l|'/|l| - \sigma z^2/m$; on a coast it is $|l|'/|l|$. During boost $|l|' < 0$, so z decreases strictly through $\mathsf{U}_+, \mathsf{B}_+, \mathsf{C}$, with initial truncations allowed. On braking $|l|' = n > 0$. At $z = 1/2$ both the coast and boundary vector fields have $z' > 0$. Thus $\mathsf{C} \to \mathsf{B}_-$ is regular and cannot return to the tie: every encounter with that level would point upward. No sliding arc or terminal m_6 event is available.

At $z = 1$ both powered fields give $z'/z = n/|l| - \sigma/m$. The quantity $k = mn/(\sigma|l|)$ decreases strictly throughout braking, since m decreases and $|l|$ increases. An upward crossing $\mathsf{B}_- \to \mathsf{U}_-$ can occur only while $k > 1$, and a subsequent downward crossing $\mathsf{U}_- \to \mathsf{B}_-$ only when $k < 1$; no further reversal is possible. Consequently every such finite-switch normal candidate has the ordered pattern

$$\mathsf{U}_+\ \mathsf{B}_+\ \mathsf{C}\ \mathsf{B}_-\ \mathsf{U}_-\ \mathsf{B}_-, \tag{90}$$

with absent intervals and endpoint truncations permitted. In particular the last boundary burn ends at the endpoint, not at a return to coast. This is a ceiling-only one-dimensional result. For the two-sided 50–250 km/s engine, the 50 km/s lower exhaust limit caps acceleration and introduces additional arcs.

To state precisely what remains unresolved here, write $V_- = 50$ km/s, $V_+ = 250$ km/s, $\rho_V = V_+/V_- = 5$, and define z with $V = V_+$. For $C_m > 0$, the dimensional selector becomes

$$|a| = \begin{cases} 0, & z < 1/2, \\ 2P/(mV_+), & 1/2 < z < 1, \\ P|l|/C_m, & 1 < z < \rho_V, \\ 2P/(mV_-), & z > \rho_V. \end{cases} \tag{91}$$

The last region is a maximum-acceleration (a_+) arc, denoted $\mathsf{A}_\pm$ according to thrust sign. Its state and mass-adjoint maps are the fixed-exhaust maps above with $V = V_-$, and

$$\frac{z'}{z} = \frac{|l|'}{|l|} + \frac{\sigma_-}{m}\left(1 - \frac{2z}{\rho_V}\right), \qquad \sigma_- = 2P/V_-^2. \tag{92}$$

Thus the selector and propagation are available for both boundaries, but a complete two-sided switching-topology theorem, including all $\mathsf{A}_\pm$ visits and admissible endpoint truncations,

is left open, as are gravity, abnormal extremals and other endpoint conditions.

C. An endpoint reduction that uses the correct adjoint

This reduction and its numerical fixture retain the ceiling-only assumptions of the proved topology. Use homogeneity to fix $n = 1$. The two shooting unknowns are t_z and $\lambda_m(0) > 0$. Propagate the exact interior, boundary and coast maps, switching with the current z, until the first exhaustion $m = m_d$. Call that time $T(t_z, \lambda_m(0))$. Solve

$$x(T; t_z, \lambda_m(0)) = L, \qquad v(T; t_z, \lambda_m(0)) = 0. \tag{93}$$

Reject trajectories without exhaustion or with invalid adjoint sign or switch inequalities. Alternatively use T as a third unknown and add $m(T) = m_d$ as a third residual. Normalize the free-time Hamiltonian afterward if a positive normal scaling exists, and compare competing solutions, since the reduction does not imply uniqueness.

For $P = 1/2$, $V = 2$, $m_0 = 1$, $m_d = 1/2$, $L = 1$, independent numerical integration of the corrected selector gives $t_z = 1.22365494$, $\lambda_m(0) = 0.61524660$, $T = 2.35199587$; the maximum endpoint residual is $9.6\,10^{-8}$. The coast is approximately $[0.80144, 1.64581]$. Direct transcription with piecewise-constant acceleration, exact $(x, v, 1/m)$ propagation and the ceiling imposed at each burn cell's final mass gives:

Burn cells	Time	Endpoint error
12	2.35489579	$2.6\,10^{-12}$
24	2.35338380	$1.4\,10^{-12}$
48	2.35267314	$1.3\,10^{-12}$

The within-cell ceiling holds because inverse mass increases monotonically and the acceleration is constant. Ceiling violations at optimizer tolerance are below $9\,10^{-13}$. The transcription independently checks the trajectory equations and approaches the shooting time from above.

Appendix B
Power-System Mass Closure for the Free-Space Benchmark

The benchmark result (30) was derived for prescribed initial and dry masses. For high-power spacecraft, however, the dry mass may itself depend on the propulsion system, because both the power-conversion hardware and the radiators must be carried together with the payload. This appendix shows how the free-space minimum-time law is modified under a simple power-system mass closure.

Let P denote the useful jet power appearing in the rocket model, and let $P_{\rm tot}$ be the total onboard engine power. Define

$$\alpha_{\rm eng} \equiv \frac{P_{\rm tot}}{m_{\rm eng}}, \qquad \phi \equiv \frac{P_{\rm heat}}{P_{\rm tot}}, \qquad \rho_{\rm rad} \equiv \frac{P_{\rm heat}}{m_{\rm rad}}, \tag{94}$$

where $\alpha_{\rm eng}$ is the engine specific power, ϕ is the waste-heat fraction, and $\rho_{\rm rad}$ is the radiator specific heat-rejection capability. Since only the fraction $1 - \phi$ of the total power is converted into useful jet power,

$$P = (1-\phi)P_{\rm tot}, \qquad P_{\rm heat} = \phi P_{\rm tot} = \frac{\phi}{1-\phi}P. \tag{95}$$

Therefore,

$$m_{\rm eng} = \frac{P}{(1-\phi)\alpha_{\rm eng}}, \qquad m_{\rm rad} = \frac{\phi}{1-\phi}\frac{P}{\rho_{\rm rad}}. \tag{96}$$

Let $m_{\rm pay}$ be the payload mass. The dry mass can then be written in the affine form

$$m_{\rm dry} = m_{\rm pay} + m_{\rm eng} + m_{\rm rad} = m_{\rm pay} + \beta P, \tag{97}$$

with

$$\beta \equiv \frac{1}{(1-\phi)\alpha_{\rm eng}} + \frac{\phi}{(1-\phi)\rho_{\rm rad}}. \tag{98}$$

The quantity β has units of kg/W and is the inverse of the effective propulsion-system specific power,

$$\alpha_{\rm sys} \equiv \beta^{-1}. \tag{99}$$

To close the mass model, assume a fixed fuel-to-dry-mass ratio

$$\mathcal{R} \equiv \frac{m_{\rm fuel}}{m_{\rm dry}}. \tag{100}$$

Then

$$m_i = (1+\mathcal{R})m_{\rm dry}, \tag{101}$$

and the parameter Δ from (30) becomes

$$\Delta = \frac{1}{m_{\rm dry}} - \frac{1}{m_i} = \frac{\mathcal{R}}{(1+\mathcal{R})m_{\rm dry}}. \tag{102}$$

Substituting (97) and (102) into the benchmark law (30) yields

$$T(P) = \left[\frac{6L^2(1+\mathcal{R})}{\mathcal{R}}\left(\frac{m_{\rm pay}}{P} + \beta\right)\right]^{1/3}. \tag{103}$$

Equation (103) is the free-space minimum-time law when the dry mass is allowed to scale linearly with propulsion power.

Several immediate consequences follow. First,

$$T^3(P) = \frac{6L^2(1+\mathcal{R})}{\mathcal{R}}\left(\frac{m_{\rm pay}}{P} + \beta\right), \tag{104}$$

so that

$$\frac{d}{dP}T^3(P) = -\frac{6L^2(1+\mathcal{R})}{\mathcal{R}}\frac{m_{\rm pay}}{P^2} < 0. \tag{105}$$

Under the idealized assumptions of this closure—unlimited exhaust speed, fixed waste-heat fraction ϕ, fixed engine specific power $\alpha_{\rm eng}$, and fixed radiator specific heat rejection $\rho_{\rm rad}$, all independent of operating point—the transfer time decreases monotonically with power and has no finite mathematical optimum:

$$P^\star \to \infty. \tag{106}$$

This formal limit should not be interpreted as a physically achievable design target, since at very large P the assumed linear mass closure breaks down: in practice $\rho_{\rm rad}$ depends on the radiator operating temperature, $\alpha_{\rm eng}$ is bounded by materials and shielding, and the exhaust speed itself is bounded above. The relevant practical implication is rather the second observation: the family approaches the asymptotic lower bound

$$T_\infty = \left[\frac{6L^2(1+\mathcal{R})}{\mathcal{R}}\,\beta\right]^{1/3}. \tag{107}$$

This lower bound arises because increasing P also increases the dry mass that must be accelerated.

A convenient engineering scale is obtained by equating propulsion-hardware mass to payload mass,

$$m_{\rm eng} + m_{\rm rad} = m_{\rm pay}. \tag{108}$$

Using (97), this defines the characteristic power

$$\beta P_{\rm knee} = m_{\rm pay}, \qquad P_{\rm knee} = \frac{m_{\rm pay}}{\beta} = \alpha_{\rm sys} m_{\rm pay}. \tag{109}$$

At this point the dry mass is $m_{\rm dry} = 2m_{\rm pay}$, and (103) can be written as

$$T(P) = T_\infty \left(1 + \frac{P_{\rm knee}}{P}\right)^{1/3}. \tag{110}$$

Hence

$$T(P_{\rm knee}) = 2^{1/3} T_\infty. \tag{111}$$

Equations (103)–(111) provide an engineering interpretation of the free-space benchmark. They are not used in the subsequent Sun-field optimal-control formulation, but they show how the cubic-root time law is modified when propulsion-system mass scales with power.

APPENDIX C
DIRECT METHOD USING SPLINE PARAMETERIZATION

Before adopting the indirect method based on Pontryagin's Maximum Principle (PMP) detailed in the main body of this paper, a more straightforward *direct method* was explored. Direct methods seek to find the optimal trajectory by parameterizing the path with a set of functions and then using a nonlinear programming solver to find the best parameters. This approach avoids the complexity of deriving and solving the costate equations.

Our implementation parameterized the spacecraft's trajectory, $\mathbf{r}(t) = (x(t), y(t))$, using cubic splines. The core idea is to define the path by a small number of control points, which become the variables for optimization.

The procedure was as follows:

1. A fixed number of N control points, $(\mathbf{p}_i, t_i)$ for $i = 0, \ldots, N-1$, were defined along the time axis of the transfer. The endpoints, $\mathbf{p}_0$ and $\mathbf{p}_{N-1}$, were fixed to match the start and end positions of the transfer arc. The initial and final velocities were enforced by connecting the transfer arc to the initial and final circular orbits.
2. The positions of the intermediate $N-2$ control points were the free variables for the optimization.
3. For any given set of control points, two cubic splines, $x_s(t)$ and $y_s(t)$, were fitted to create a continuous and smooth trajectory $\mathbf{r}(t) = (x_s(t), y_s(t))$. A cubic spline is defined piecewise by third-order polynomials, ensuring that the first and second derivatives (velocity and acceleration) are continuous across the entire path.
4. The required velocity $\dot{\mathbf{r}}(t)$ and acceleration $\ddot{\mathbf{r}}(t)$ vectors were obtained by numerically differentiating the spline functions.
5. With the total acceleration $\ddot{\mathbf{r}}(t)$ known, the required thrust acceleration was calculated by subtracting the gravitational component:

$$\mathbf{a}_T(t) = \ddot{\mathbf{r}}(t) - \mathbf{a}_g(\mathbf{r}(t)). \tag{112}$$

6. The instantaneous mass flow rate was then determined from the thrust acceleration magnitude using (2). Integrating this rate backwards in time yielded the total propellant consumed for the maneuver.
7. A nonlinear optimizer (specifically, the Nelder–Mead method) was used to adjust the positions of the intermediate control points to minimize a cost function based on the final fuel expenditure and flight time.

Limitations and Stagnation in Local Minima

While conceptually simple, this direct method proved inadequate for finding high-performance trajectories. The primary difficulty was the nature of the optimization landscape. The relationship between the position of a control point and the final mission cost is highly complex and nonlinear. As a result, the search space is filled with a vast number of local minima.

For a very small number of intermediate control points ($N = 3$ or $N = 4$ total), the method could find a reasonable, smooth trajectory. However, as the number of control points was increased to grant the path more freedom and potential for higher performance, the dimensionality of the search space grew, and the solver would invariably become trapped in a suboptimal local minimum.

An attempt was made to mitigate this by using a continuation strategy on the number of control points: first, an optimal path with $N = 4$ points was found. Then, a fifth point was added at the location of maximum curvature, and the system was re-optimized. While this yielded marginal improvements, the solution quality consistently stagnated far below that achievable with the PMP-based method. The direct spline method lacks the rigorous guidance provided by the costate (or "primer vector") variables, which point the way toward optimality across the entire trajectory. These difficulties ultimately motivated the adoption of the indirect shooting method presented in this work.

REFERENCES

[1] B. A. Conway, ed., *Spacecraft Trajectory Optimization*. Cambridge, UK: Cambridge University Press, 2010.

[2] B. P. Pino and K. C. Howell, "An energy-informed adaptive algorithm for low-thrust spacecraft cislunar trajectory design," in *AAS/AIAA Space Flight Mechanics Meeting*, 2020. AAS 20-579.

[3] J. A. Fogel, M. Widner, J. Williams, and A. Batcha, "Multi-impulse to time optimal finite burn trajectory conversion," tech. rep., NASA Johnson Space Center, 2020.

[4] S. Mazouffre, "Electric propulsion for satellites and spacecraft: Established technologies and novel approaches," *Plasma Sources Science and Technology*, vol. 25, no. 3, p. 033002, 2016.

[5] A. Peakman and B. Lindley, "A review of nuclear electric fission space reactor technologies for achieving high-power output and operating with HALEU fuel," *Progress in Nuclear Energy*, vol. 163, p. 104815, 2023.

[6] J. H. Scott, "The value proposition of multi-megawatt electric power/propulsion for the human exploration of mars," in *International Astronautical Conference*, 2019. JSC-E-DAA-TN73357.

[7] F. R. C. Díaz, J. A. Carr, L. Johnson, W. Johnson, G. Genta, and P. F. Maffione, "Solar electric propulsion for human mars missions," *Acta Astronautica*, vol. 160, pp. 183–194, 2019.

[8] T. F. Sheerin, E. Petro, K. Winters, P. Lozano, and P. Lubin, "Fast solar system transportation with electric propulsion powered by directed energy," *Acta Astronautica*, vol. 179, pp. 78–87, 2021.

[9] K. E. Tsiolkovsky, "Exploration of cosmic space by means of reaction devices," 1903. English translation, NASA TT F-243, 1965.

[10] E. A. Bering, B. W. Longmier, M. G. Ballenger, C. S. Olsen, J. P. Squire, and F. R. C. Díaz, "Performance studies of the VASIMR VX-200," in *49th AIAA Aerospace Sciences Meeting*, (Orlando, FL), 2011.

[11] G. D. Pasquale, M. Sanjurjo-Rivo, and D. P. Grande, "A multi-objective variable-specific impulse maneuvers optimization methodology," *Advances in Space Research*, vol. 73, no. 1, pp. 20–30, 2024.

[12] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*. New York, NY: Wiley, 9 ed., 2017.

[13] L. S. Pontryagin, V. G. Boltyanskii, R. V. Gamkrelidze, and E. F. Mishchenko, *The Mathematical Theory of Optimal Processes*. Oxford, UK: Pergamon Press, 1962.

[14] R. Storn and K. Price, "Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, no. 4, pp. 341–359, 1997.

[15] E. L. Allgower and K. Georg, *Numerical Continuation Methods: An Introduction*. Berlin, Germany: Springer, 1990.

[16] M. J. D. Powell, "An efficient method for finding the minimum of a function of several variables without calculating derivatives," *The Computer Journal*, vol. 7, pp. 155–162, 1964.

[17] J. Olšina, "Minimum-effort planning for constant-power spacecraft: Timing, engine limits and planetary encounters," 2026. Revised manuscript, 29 October 2026; distributed in the project archive.

[18] J. Olšina, "Time-optimal heliocentric transfers with a constant-power, variable-isp engine," 2025. DOI: 10.5281/zenodo.17521033. Code and data: https://github.com/irigi/VariableISPRocketTrajectories.

[19] H. Oberth, *Die Rakete zu den Planetenräumen*. Munich, Germany: R. Oldenbourg, 1923. English translation: The Rocket into Planetary Space.

[20] R. F. Hartl, S. P. Sethi, and R. G. Vickson, "A survey of the maximum principles for optimal control problems with state constraints," *SIAM Review*, vol. 37, no. 2, pp. 181–218, 1995.

[21] J. Olšina, "Pursuit, rendezvous, and evasion with constant-power, variable-isp rockets," 2026. Corrected companion manuscript, RocketPursuitEvasion.tex, distributed with this revision.

TABLE I

TRANSFER TIME IN DAYS FROM THE ORIGINAL SOLVER (U: UNRESOLVED) FROM A CIRCULAR ORBIT OF 1 AU AND FINAL POLAR ANGLE (AT ROCKET ARRIVAL).

Polar angle	**Target Radius (AU)**						
(°)	0.390	0.720	1.520	5.200	9.580	19.220	30.050
0	U	U	U	220.50	337.82	539.55	724.65
5	U	U	84.99	214.66	331.66	532.47	716.76
10	U	U	73.01	208.91	325.52	525.31	708.75
15	U	U	60.88	203.32	319.44	518.17	700.69
20	U	U	51.35	197.98	313.53	511.09	692.64
25	U	U	45.94	192.95	307.83	504.13	684.71
30	U	32.66	43.80	188.35	302.40	497.40	676.93
35	U	29.03	43.63	184.23	297.29	490.93	669.37
40	U	28.92	44.61	180.59	292.59	484.76	662.08
45	U	29.86	46.15	177.47	288.32	478.95	655.19
50	U	31.19	48.07	174.94	284.48	473.58	648.68
55	U	32.74	50.23	172.94	281.10	468.66	642.60
60	U	34.42	52.55	171.44	278.25	464.20	637.00
65	U	36.18	54.96	170.42	275.88	460.21	631.92
70	U	38.00	57.46	169.90	273.97	456.76	627.37
75	U	39.84	59.98	169.80	272.50	453.79	623.38
80	U	41.69	62.51	170.05	271.54	451.39	619.93
85	U	43.55	65.05	170.65	270.98	449.42	617.06
90	46.97	45.39	67.58	171.57	270.81	447.98	614.72
95	45.78	47.22	70.07	172.77	270.99	447.01	612.89
100	45.47	49.04	72.55	174.17	271.54	446.46	611.58
105	45.54	50.82	74.97	175.77	272.38	446.33	610.73
110	45.86	52.58	77.35	177.53	273.50	446.57	610.37
115	46.37	54.30	79.68	179.42	274.83	447.19	610.43
120	47.00	55.99	81.93	181.41	276.38	448.11	610.89
125	47.72	57.63	84.12	183.48	278.09	449.29	611.70
130	48.49	59.22	86.24	185.58	279.93	450.73	612.82
135	49.31	60.76	88.27	187.70	281.87	452.36	614.21
140	50.16	62.24	90.22	189.81	283.87	454.17	615.83
145	51.02	63.68	92.07	191.90	285.92	456.10	617.64
150	51.89	65.05	93.83	193.93	287.97	458.09	619.57
155	52.75	66.36	95.50	195.89	289.99	460.14	621.61
160	53.61	67.61	97.05	197.76	291.96	462.19	623.69
165	54.46	68.79	98.50	199.53	293.86	464.22	625.79
170	55.28	69.89	99.84	201.18	295.66	466.18	627.86
175	56.08	70.94	101.08	202.72	297.35	468.07	629.88
180	56.86	71.91	102.22	204.12	298.92	469.85	631.81
185	57.61	72.83	103.25	205.38	300.34	471.50	633.63
190	58.34	73.67	104.18	206.51	301.63	473.03	635.31
195	59.03	74.46	105.02	207.53	302.79	474.41	636.87
200	59.68	75.17	105.75	208.39	303.79	475.63	638.27
205	60.31	75.83	106.40	209.13	304.67	476.72	639.53
210	60.90	76.42	106.97	209.77	305.42	477.67	640.64
215	61.45	76.96	107.46	210.31	306.05	478.48	641.61
220	61.97	77.44	107.86	210.74	306.57	479.17	642.45
225	62.47	77.86	108.20	211.07	306.99	479.74	643.16
230	62.93	78.23	108.48	211.33	307.32	480.22	643.76
235	63.35	78.56	108.69	211.52	307.57	480.60	644.27
240	63.74	78.84	108.85	211.65	307.75	480.91	644.68
245	64.10	79.07	108.95	211.72	307.87	481.14	645.02
250	64.44	79.26	109.01	211.75	307.94	481.31	645.29
255	64.74	79.42	109.03	211.73	307.97	481.43	645.51
260	65.02	79.54	109.01	211.68	307.96	481.51	645.68
265	65.26	79.62	108.96	211.60	307.92	481.57	645.82
270	65.48	79.66	108.88	211.50	307.87	481.59	645.94
275	65.69	79.69	108.78	211.40	307.79	481.61	646.04
280	65.86	79.69	108.66	211.27	307.72	481.62	646.12
285	66.02	79.65	108.52	211.14	307.63	481.62	646.22
290	66.16	79.61	108.37	211.00	307.55	481.64	646.31
295	66.27	79.54	108.21	210.88	307.48	481.67	646.43
300	66.38	79.47	108.03	210.75	307.42	481.71	646.57
305	66.47	79.37	107.87	210.65	307.38	481.79	646.74
310	66.56	79.27	107.70	210.56	307.37	481.89	646.96
315	66.63	79.18	107.54	210.49	307.38	482.04	647.22
320	66.70	79.07	107.39	210.45	307.43	482.23	647.54
325	66.77	78.98	107.26	210.43	307.52	482.48	647.92
330	66.84	78.88	107.14	210.45	307.64	482.79	648.38
335	66.90	78.81	107.03	210.51	307.82	483.16	648.91
340	66.99	78.74	106.97	210.61	308.06	483.61	649.54
345	67.08	78.70	106.94	210.75	308.36	484.14	650.27
350	67.19	78.70	106.94	210.96	308.71	484.76	651.11
355	67.32	78.73	107.00	211.22	309.15	485.47	652.06
360	67.49	78.81	107.10	211.54	309.66	486.29	653.13

TABLE II

TRANSFER TIME IN DAYS FROM THE ORIGINAL SOLVER (U: UNRESOLVED) FROM A CIRCULAR ORBIT OF 1 AU AND INITIAL POLAR ANGLE (AT ROCKET DEPARTURE).

Polar Angle	**Target Radius (AU)**						
(°)	0.390	0.720	1.520	5.200	9.580	19.220	30.050
-100	46.96	U	U	281.19	401.02	604.63	U
-95	46.26	U	U	279.72	400.45	605.12	U
-90	45.76	U	U	277.93	399.55	605.20	U
-85	45.48	U	U	275.79	398.22	604.85	U
-80	45.50	U	U	273.34	396.54	604.07	U
-75	45.92	U	U	270.57	394.49	602.85	U
-70	47.06	U	U	267.50	392.06	601.12	U
-65	49.08	U	U	264.11	389.20	598.91	U
-60	51.87	U	U	260.40	385.96	596.20	U
-55	54.65	U	U	256.39	382.34	592.99	783.40
-50	56.90	U	U	252.12	378.37	589.31	779.93
-45	58.68	U	U	247.60	374.05	585.12	775.84
-40	60.08	U	85.30	242.86	369.37	580.49	771.21
-35	61.19	U	80.06	237.92	364.40	575.39	766.05
-30	62.11	36.38	74.72	232.82	359.17	569.87	760.36
-25	62.88	33.69	69.37	227.58	353.71	564.00	754.17
-20	63.51	31.65	64.02	222.28	348.06	557.81	747.56
-15	64.05	29.77	58.87	216.96	342.25	551.34	740.58
-10	64.50	28.86	53.88	211.67	336.37	544.62	733.26
-5	64.89	29.30	49.74	206.46	330.45	537.74	725.67
0	65.21	31.38	46.38	201.41	324.55	530.74	717.88
5	65.48	34.85	44.28	196.58	318.72	523.70	709.96
10	65.72	39.16	43.59	192.04	313.06	516.69	701.98
15	65.92	43.78	44.17	187.83	307.58	509.73	694.01
20	66.09	48.26	45.84	184.00	302.34	502.93	686.11
25	66.23	52.44	48.21	180.58	297.40	496.34	678.37
30	66.35	56.21	51.07	177.61	292.82	490.00	670.84
35	66.45	59.57	54.25	175.13	288.65	483.96	663.57
40	66.55	62.47	57.60	173.15	284.87	478.29	656.62
45	66.63	65.02	61.04	171.63	281.52	473.04	650.08
50	66.70	67.23	64.51	170.57	278.62	468.22	643.97
55	66.77	69.13	67.95	169.94	276.22	463.83	638.30
60	66.84	70.78	71.31	169.78	274.27	459.92	633.10
65	66.91	72.19	74.60	169.99	272.76	456.56	628.48
70	67.01	73.43	77.78	170.54	271.68	453.63	624.33
75	67.11	74.48	80.81	171.40	271.05	451.26	620.80
80	67.24	75.39	83.70	172.59	270.82	449.33	617.74
85	67.41	76.18	86.43	174.01	270.94	447.93	615.27
90	67.62	76.85	88.98	175.63	271.40	446.97	613.30
95	67.90	77.44	91.37	177.43	272.19	446.44	611.88
100	68.28	77.91	93.55	179.39	273.28	446.33	610.94
105	68.79	78.33	95.58	181.45	274.59	446.59	610.43
110	69.52	78.67	97.41	183.58	276.10	447.23	610.40
115	U	78.96	99.09	185.77	277.82	448.16	610.75
120	U	79.18	100.57	187.96	279.67	449.36	611.45
125	U	79.36	101.92	190.15	281.62	450.81	612.49
130	U	79.50	103.10	192.30	283.64	452.46	613.81
135	U	79.59	104.14	194.39	285.71	454.29	615.39
140	U	79.65	105.06	196.39	287.79	456.23	617.15
145	U	79.68	105.84	198.29	289.84	458.24	619.06
150	U	79.69	106.53	200.09	291.84	460.29	621.07
155	U	79.66	107.11	201.74	293.77	462.35	623.15
160	U	79.63	107.59	203.26	295.60	464.39	625.26
165	U	79.57	107.99	204.64	297.31	466.36	627.35
170	U	79.49	108.32	205.88	298.90	468.25	629.38
175	U	79.41	108.58	206.97	300.33	470.02	631.35
180	U	79.32	108.76	207.93	301.64	471.67	633.19
185	U	79.22	108.90	208.75	302.80	473.19	634.91
190	U	79.12	108.99	209.46	303.81	474.55	636.50
195	U	79.02	109.02	210.05	304.69	475.76	637.95
200	U	78.93	109.02	210.53	305.44	476.84	639.25
205	U	78.85	109.00	210.92	306.07	477.78	640.39
210	U	78.78	108.93	211.22	306.59	478.57	641.39
215	U	78.72	108.84	211.44	307.00	479.25	642.26
220	U	78.70	108.73	211.59	307.33	479.80	643.01
225	U	78.71	108.60	211.69	307.58	480.28	643.63
230	U	78.76	108.46	211.74	307.76	480.64	644.16
235	U	78.87	108.31	211.74	307.88	480.94	644.60
240	U	79.06	108.14	211.71	307.95	481.16	644.95
245	U	79.33	107.98	211.64	307.97	481.33	645.23
250	U	79.74	107.81	211.55	307.96	481.44	645.46
255	U	80.32	107.65	211.45	307.92	481.52	645.64
260	U	81.16	107.49	211.33	307.86	481.57	645.79

TABLE III

TRANSFER TIME IN DAYS FROM THE ORIGINAL SOLVER (U: UNRESOLVED) TO A CIRCULAR ORBIT OF 1 AU AND FINAL POLAR ANGLE (AT ROCKET ARRIVAL).

Polar Angle	**Origin Radius (AU)**						
(°)	0.390	0.720	1.520	5.200	9.580	19.220	30.050
0	U	U	95.77	220.50	337.82	539.56	724.64
5	U	U	84.93	214.67	331.66	532.46	716.76
10	U	U	73.02	208.91	325.51	525.31	708.76
15	U	U	60.90	203.30	319.44	518.16	700.69
20	U	U	51.21	197.97	313.53	511.08	692.65
25	U	U	45.92	192.97	307.82	504.14	684.70
30	U	32.29	43.85	188.36	302.39	497.40	676.91
35	U	28.90	43.65	184.19	297.29	490.93	669.34
40	U	28.92	44.59	180.58	292.60	484.76	662.08
45	U	29.84	46.15	177.50	288.32	478.95	655.19
50	U	31.18	48.07	174.95	284.47	473.58	648.68
55	U	32.73	50.23	172.91	281.10	468.66	642.61
60	U	34.42	52.55	171.44	278.26	464.19	637.03
65	U	36.19	54.97	170.45	275.88	460.24	631.95
70	U	37.99	57.46	169.91	273.97	456.77	627.37
75	U	39.83	59.98	169.77	272.50	453.80	623.40
80	U	41.69	62.51	170.05	271.54	451.38	619.96
85	49.67	43.54	65.05	170.67	270.98	449.42	617.07
90	46.93	45.39	67.57	171.58	270.81	447.98	614.71
95	45.83	47.22	70.08	172.74	270.99	447.01	612.86
100	45.46	49.03	72.54	174.16	271.54	446.45	611.58
105	45.51	50.82	74.97	175.77	272.38	446.33	610.73
110	45.85	52.58	77.36	177.54	273.50	446.58	610.38
115	46.37	54.31	79.67	179.42	274.83	447.17	610.46
120	47.00	55.99	81.93	181.41	276.38	448.10	610.91
125	47.72	57.63	84.12	183.47	278.09	449.30	611.70
130	48.49	59.22	86.23	185.58	279.93	450.74	612.82
135	49.31	60.76	88.27	187.70	281.87	452.36	614.21
140	50.15	62.24	90.22	189.81	283.87	454.17	615.83
145	51.02	63.68	92.07	191.90	285.92	456.10	617.64
150	51.89	65.04	93.84	193.93	287.97	458.09	619.58
155	52.75	66.35	95.49	195.89	289.99	460.14	621.61
160	53.61	67.60	97.05	197.76	291.96	462.19	623.69
165	54.45	68.79	98.49	199.53	293.86	464.22	625.79
170	55.28	69.89	99.84	201.18	295.66	466.18	627.86
175	56.08	70.94	101.08	202.72	297.35	468.07	629.88
180	56.86	71.91	102.23	204.13	298.92	469.85	631.81
185	57.61	72.83	103.26	205.39	300.35	471.50	633.62
190	58.33	73.68	104.18	206.52	301.64	473.03	635.31
195	59.02	74.46	105.02	207.52	302.79	474.40	636.87
200	59.68	75.17	105.76	208.39	303.80	475.63	638.27
205	60.31	75.83	106.41	209.14	304.67	476.72	639.52
210	60.90	76.42	106.97	209.77	305.42	477.67	640.64
215	61.46	76.96	107.45	210.30	306.06	478.48	641.61
220	61.98	77.44	107.87	210.73	306.58	479.17	642.44
225	62.47	77.86	108.21	211.07	306.99	479.75	643.17
230	62.93	78.23	108.48	211.34	307.32	480.22	643.76
235	63.35	78.57	108.69	211.52	307.58	480.60	644.27
240	63.75	78.84	108.84	211.65	307.75	480.90	644.68
245	64.10	79.07	108.95	211.72	307.87	481.14	645.02
250	64.44	79.27	109.01	211.75	307.94	481.31	645.30
255	64.74	79.41	109.03	211.73	307.97	481.43	645.51
260	65.01	79.53	109.01	211.68	307.96	481.51	645.68
265	65.26	79.62	108.97	211.61	307.92	481.57	645.82
270	65.49	79.66	108.88	211.51	307.87	481.60	645.93
275	65.69	79.68	108.78	211.39	U	481.61	646.03
280	65.86	79.68	108.66	211.26	U	481.62	646.12
285	66.02	79.65	108.52	211.14	U	481.62	646.21
290	66.16	79.61	108.36	211.00	U	481.63	646.31
295	66.27	79.54	108.21	210.88	U	481.66	646.43
300	66.38	79.47	108.04	210.75	U	481.71	646.57
305	66.47	79.37	107.87	210.65	U	481.78	646.75
310	66.56	79.27	107.70	210.56	U	481.89	646.96
315	66.63	79.17	107.54	210.49	U	482.04	647.22
320	66.70	79.07	107.39	210.44	U	482.24	647.54
325	66.76	78.97	107.25	210.43	U	482.48	647.93
330	66.83	78.89	107.13	210.45	U	482.79	648.38
335	66.90	78.80	107.03	210.51	U	483.16	648.91
340	66.99	78.74	106.97	210.60	U	483.61	649.55
345	67.08	78.70	106.94	210.75	U	484.14	650.27
350	67.19	78.70	106.94	210.96	U	484.76	651.11
355	67.32	78.73	107.00	211.22	U	485.47	652.06
360	67.48	78.82	107.10	211.54	U	486.28	653.13

TABLE IV

TRANSFER TIME IN DAYS FROM THE ORIGINAL SOLVER (U: UNRESOLVED) TO A CIRCULAR ORBIT OF 1 AU AND INITIAL POLAR ANGLE (AT ROCKET DEPARTURE).

Polar Angle	**Origin Radius (AU)**						
(°)	0.390	0.720	1.520	5.200	9.580	19.220	30.050
-100	46.94	U	137.22	281.20	401.02	604.64	789.45
-95	46.28	U	133.71	279.71	400.45	605.10	790.72
-90	45.79	U	130.04	277.91	399.53	605.19	791.59
-85	45.47	U	126.23	275.81	398.24	604.86	792.02
-80	45.48	U	122.29	273.35	396.55	604.09	791.99
-75	45.94	U	118.16	270.58	394.50	602.83	791.40
-70	47.06	U	113.88	267.49	392.03	601.11	790.28
-65	49.09	U	109.49	264.09	389.19	598.93	788.57
-60	51.87	U	104.93	260.40	385.96	596.20	786.30
-55	54.64	U	100.21	256.39	382.37	593.01	783.38
-50	56.91	U	95.40	252.12	378.37	589.32	779.89
-45	58.67	U	90.40	247.61	374.03	585.12	775.84
-40	60.08	42.22	85.27	242.85	369.36	580.46	771.21
-35	61.20	39.24	80.06	237.91	364.41	575.38	766.07
-30	62.11	36.39	74.72	232.81	359.17	569.87	760.34
-25	62.88	33.74	69.36	227.59	353.71	564.01	754.16
-20	63.51	31.37	64.05	222.28	348.05	557.81	747.56
-15	64.05	29.82	58.75	216.95	342.25	551.33	740.58
-10	64.50	28.85	53.84	211.66	336.37	544.62	733.26
-5	64.88	29.29	49.74	206.47	330.45	537.73	725.67
0	65.21	31.36	46.31	201.42	324.55	530.74	717.89
5	65.49	34.83	44.22	196.60	318.73	523.70	709.96
10	65.72	39.16	43.58	192.04	313.06	516.67	701.98
15	65.92	43.77	44.15	187.81	307.57	509.73	694.00
20	66.09	48.26	45.83	183.96	302.33	502.94	686.10
25	66.23	52.45	48.22	180.57	297.40	496.34	678.35
30	66.35	56.21	51.07	177.63	292.84	490.00	670.82
35	66.46	59.57	54.25	175.16	288.65	483.95	663.55
40	66.54	62.47	57.61	173.14	284.87	478.29	656.62
45	66.63	65.02	61.04	171.60	281.51	473.04	650.08
50	66.70	67.22	64.50	170.57	278.64	468.22	643.97
55	66.77	69.13	67.94	169.97	276.22	463.84	638.28
60	66.84	70.77	71.32	169.78	274.27	459.96	633.13
65	66.92	72.19	74.61	169.97	272.76	456.55	628.48
70	67.01	73.42	77.78	170.54	271.68	453.63	624.35
75	67.12	74.48	80.81	171.43	271.05	451.26	620.77
80	67.24	75.39	83.70	172.59	270.82	449.33	617.75
85	67.41	76.18	86.42	173.99	270.94	447.93	615.28
90	67.62	76.85	88.98	175.64	271.40	446.97	613.33
95	U	77.43	91.36	177.44	272.19	446.44	611.88
100	U	77.92	93.56	179.39	273.28	446.33	610.94
105	U	78.33	95.57	181.45	274.59	446.60	610.43
110	U	78.67	97.42	183.58	276.10	447.20	610.38
115	U	78.95	99.07	185.77	277.82	448.15	610.75
120	U	79.18	100.57	187.97	279.67	449.37	611.47
125	U	79.36	101.92	190.15	281.62	450.82	612.50
130	U	79.49	103.11	192.30	283.64	452.46	613.81
135	U	79.59	104.14	194.39	285.71	454.29	615.39
140	U	79.65	105.06	196.39	287.79	456.23	617.15
145	U	79.68	105.85	198.29	289.84	458.24	619.06
150	U	79.69	106.54	200.09	291.84	460.29	621.08
155	U	79.66	107.11	201.74	293.77	462.35	623.15
160	U	79.63	107.60	203.26	295.60	464.39	625.26
165	U	79.57	108.00	204.64	297.31	466.36	627.35
170	U	79.49	108.32	205.88	298.90	468.25	629.38
175	U	79.41	108.58	206.99	300.34	470.02	631.35
180	U	79.32	108.76	207.93	301.65	471.67	633.20
185	U	79.22	108.90	208.75	302.80	473.19	634.91
190	U	79.12	108.98	209.46	303.82	474.54	636.50
195	U	79.02	109.02	210.05	304.69	475.76	637.95
200	U	78.93	109.03	210.52	305.45	476.84	639.25
205	U	78.85	108.99	210.92	306.08	477.77	640.40
210	U	78.77	108.93	211.22	306.59	478.57	641.40
215	U	78.72	108.84	211.44	307.01	479.25	642.26
220	U	78.70	108.73	211.59	307.34	479.81	643.01
225	U	78.71	108.60	211.70	307.59	480.28	643.64
230	U	78.76	108.46	211.74	307.76	480.64	644.17
235	U	78.87	108.30	211.74	307.88	480.93	644.60
240	U	79.06	108.15	211.71	307.94	481.17	644.95
245	U	U	107.97	211.64	307.97	481.33	645.24
250	U	U	107.81	211.56	307.96	481.44	645.47
255	U	U	107.65	211.45	307.92	481.52	645.64
260	U	U	107.49	211.33	307.87	481.57	645.79